
Calibration of Local Volatility Models

A Tikhonov regularization approach to the Dupire inverse problem

written by

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ABSTRACT

Resumen en español:

Aquí ponemos el resumen del Trabajo de Fin de Máster, de unas cinco líneas.

Abstract in English:

Here we put a summary of the Master Thesis, about five lines long.

Keywords: Local volatility, Dupire equation, inverse problems, Tikhonov regularization

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INTRODUCTION

Financial products have been around for centuries, mostly in form of credit and insurance, more recently in form of shared ownership of companies. This products gave born to their natural extension, the derivatives, contracts whose value depends on the value of an underlying asset, they offer ways to transfer risk, also known as hedging, and to speculate on the future of the underlying, without owning it. For many knowing the price to buy or sell such contracts is crucial for their business, thus the more popular and more traded a product is, the more interest of modeling its price. This brings us to 1973 when coincidentally two big events for financial options, happened.

An option is a contract that gives its holder the right, but not the obligation, to buy or sell an asset at an agreed price on an agreed date. In 1973 Black and Scholes [5] published their now famous model to price European options, Merton [18] published the same result derived from a different and more general perspective. This model set the foundations to price options that coincidentally started to be listed in the CBOE, a large financial exchange in the US.

As we will see in Chapter 4, Black-Scholes model through some assumptions on how the underlying asset dynamics are, models the price of an option as a partial differential equation (PDE),

$$\frac{\partial f}{\partial t} = rf(t, S_t) - r \frac{\partial f}{\partial S} S_t - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2,$$

one assumption is that the volatility σ is constant, meaning that the variance of the underlying asset dynamics is constant in time, such simplification allows the PDE to have a closed solution,

$$f(t, S_t) = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

making the model very popular and useful in practice.

Before 1987, the empirical evidence backed the constant volatility hypothesis, but after the 1987 crash, markets start displaying what is known as the smile, volatility was not constant

anymore, it depended on the strike price of the option. The change made the Black-Scholes model inconsistent with the market, and woke up the interest of modeling the volatility as a function of strike and time. Many models have been proposed to model the volatility, we will focus on the one proposed by Dupire [10], known as the local volatility model, where the volatility is a deterministic function of time and spot price,

$$\sigma(t, S)^2 = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}} \Big|_{\substack{K=S, \\ T=t}},$$

and also models the price of an option as a PDE,

$$\frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} - \frac{\sigma^2(T, K)}{2} K^2 \frac{\partial^2 C}{\partial K^2} = -qC.$$

We will see it more in detail in Chapter 6.

In practice, all this models require a critical step to be useful, knowing the value of their parameters, in particular the volatility. Local volatility model requires to know the function describing the local volatility surface. In PDE theory this is known as an inverse problem, where the goal is to find the parameters of a PDE given some observations of its solution.

1.1 Objectives

The main goal of this thesis is to present a method to calibrate the local volatility model, while also presenting the theory behind the pricing model, going through some basics of stochastic calculus, the Black-Scholes equation and arbitrage theory. So will set a base to understand the local volatility model and its calibration as an inverse problem.

We also aim to do a numerical implementation of both Black-Scholes and local volatility models, and their calibration, to check the theory, compare them and test some of the results presented in the literature. All the code can be found in the repository <https://github.com/M315/TFM>.

1.2 Structure of the document

Chapters 2 and 3 collect the preliminary notions on probability spaces, filtrations, martingales and Brownian motion, the Itô integral, the Itô formula and stochastic differential equations.

Chapter 4 derives the Black-Scholes model, solves its PDE to obtain the pricing formula, and defines the implied volatility and its surface. There is numerical implementation of the implied volatility surface with a more detailed description in Appendix A.

Chapter 5 generalizes the pricing argument, presenting the PDE and the martingale methods for a general contingent claim and the two fundamental theorems of asset pricing.

Chapter 6 introduces the local volatility model, deriving the Dupire formula and the forward equation, states the no-arbitrage conditions on the price surface, and collects some results on the local volatility model.

Chapter 7 treats the calibration problem. It explains the ill-posedness, sets up the parameter-to-solution map and the Tikhonov functional, computes the gradient by the adjoint method, reviews the convergence rates and the source condition, and describes the discretization and the minimization. Here again we present a numerical implementation of the calibration method with a more detailed description in Appendix B.

PRELIMINARY NOTIONS

2.1 Probability theory

The foundation of modern probability theory is based on measure theory.

Definition 2.1 (σ -algebra). A σ -algebra $\mathcal{F}$ on a set Ω is a collection of subsets of Ω that

- $\Omega \in \mathcal{F}$
- if $A \in \mathcal{F}$ then $\Omega \setminus A \in \mathcal{F}$
- if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Definition 2.2 (Measure). Let Ω be a set and $\mathcal{F}$ a σ -algebra on Ω . A function $\mu : \mathcal{F} \rightarrow [0, \infty]$ is **measure** if satisfies

1. $\mu(\emptyset) = 0$
2. $\mu(A) \geq 0$ for all $A \in \mathcal{F}$
3. if $\{A_i\}_{i=1}^{\infty} \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$ for all $i \neq j$ then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

If μ is a measure on $\mathcal{F}$ such that $\mu(\Omega) = 1$, then μ is called a **probability measure**.

Definition 2.3 (Probability space). A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is the set of all possible outcomes

- $\mathcal{F}$ is a σ -algebra on Ω
- $\mathbb{P}$ is a probability measure.

For a more details in probability theory and stochastic processes, see [16] or [17].

2.2 Stochastic processes

Definition 2.4 (Stochastic processes). A stochastic process is an indexed family of random variables

$$\{X(t, \omega)\}_{t \in T},$$

on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ assuming values in $\mathbb{R}^n$.

If the set T is countable then the stochastic process is said to be *discrete*, if T is some interval of $\mathbb{R}$ then it is *continuous*.

An intuitive way to understand stochastic processes is to consider what is $X(t, \omega)$ when t or ω is fixed. If we fix $t \in T$ then the map

$$\omega \rightarrow X_t(\omega),$$

for $\omega \in \Omega$ is a random variable. If $\omega \in \Omega$ is fixed then the map

$$t \rightarrow X_\omega(t),$$

for $t \in T$ is a deterministic function, and is called *path* of $X(t, \omega)$.

Definition 2.5 (Filtration). A **filtration** is a family of σ -algebras $\{\mathcal{F}_t\}_{t \in T}$ indexed by T , such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$, where $s, t \in T$.

The idea is that a filtration $\mathcal{F}_t$ represents the information available from the past of an stochastic process up to time t . Thus we define a **filtered probability space** as a probability space with a filtration $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, \mathbb{P})$.

Definition 2.6 (Adapted process). Let X_t be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, \mathbb{P})$. The process is **adapted** to the filtration $\{\mathcal{F}_t\}_{t \in T}$ if, for every $t \in T$, the random variable X_t is $\mathcal{F}_t$ -measurable.

The idea of an adapted process is key when modeling using stochastic processes, as it ensures that the process at time t depends only on the information available up to that time, but not on future information.

Definition 2.7 (Quadratic variation [21] Ch. 8.6). Let X_t be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in T}, \mathbb{P})$. For any partition $\pi = \{t_0, t_1, \dots, t_n\} \subset [0, t]$, the π -quadratic variation of X_t is defined as the random variable

$$Q_\pi(X_t) = \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2.$$

If there is a process V_t such that $Q_\pi(X_t) \rightarrow V_t$ in probability, for any sequence of partitions π of $[0, t]$, with $\mu(\pi) \rightarrow 0$, then V_t is called the **quadratic variation** of X_t and is denoted by $\langle X \rangle_t$.

2.2.1 Martingales

A martingale is a fundamental concept in the theory of stochastic processes, formalizes the idea of a "fair game", where given all the information up to now, the best prediction for the future is simply the present value.

Definition 2.8 (Martingale). A stochastic process X_t where $t \in T$ is called a **martingale** with respect to a filtration $\{\mathcal{F}_t\}_{t \in T}$ if it satisfies the following conditions:

1. $E[|X_t|] < \infty$, for each $t \in T$
2. X_t is measurable with respect to $\mathcal{F}_t$ for each $t \in T$
3. If X_t is discrete

$$E[X_{t+1} | \mathcal{F}_t] = X_t \quad \forall t \in \mathbb{N},$$

if X_t is continuous

$$E[X_t | \mathcal{F}_s] = X_s \quad \forall s \leq t.$$

2.2.2 Brownian Motion

Brownian motion (also called Wiener process) is the fundamental stochastic process underlying continuous-time financial modeling. Formally, a Brownian motion $\{W_t\}_{(t \geq 0)}$ is defined by the following properties:

1. $W_0 = 0$
2. The map $t \rightarrow W_t(\omega)$ is almost surely continuous
3. W_t has independent increments, i.e. for $t_1 < t_2 < t_3 < t_4$, $W_{t_2} - W_{t_1}$ and $W_{t_4} - W_{t_3}$ are independent random variables
4. W_t has gaussian increments, i.e. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s \leq t$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2 .

The Brownian motion has some key features that will need afterwards, for example it can be shown that the paths $t \rightarrow W_t(\omega)$ although are almost surely continuous, they are nowhere differentiable.

Also it can be shown that the Wiener process is an almost surely continuous martingale with quadratic variation $[W, W]_t = t$. These two properties together with $W_0 = 0$ form an equivalent definition of the Wiener process, known as Lévy's characterization.

Proofs of these properties, the equivalence between the two definitions, and the existence of Brownian motion can be found in [19] and [20, Ch. 2].

STOCHASTIC CALCULUS

Most contents from this chapter are taken from [20] chapters 3 to 5 and [21].

To model the dynamics of a financial asset's value, an ordinary differential equation of the form

$$\frac{dS_t}{dt} = f(t, S_t)$$

is insufficient: it cannot capture the random fluctuations inherent in financial data. Stochastic differential equations (SDEs) add a noise term to handle this,

$$(3.1) \quad dS_t = f(t, S_t)dt + \sigma(t, S_t)dW_t,$$

where W_t is a Brownian motion. In integral form this reads

$$(3.2) \quad S_t = S_0 + \int_0^t f(s, S_s)ds + \int_0^t \sigma(s, S_s)dW_s,$$

thus a key step is to give rigorous meaning to the stochastic integral $\int \sigma dW_s$.

The most well known process that has a well defined stochastic integral is the Wiener process or Brownian motion, this stochastic integral is known as the Ito integral and will allow us to have the machinery to study SDEs and thus model the dynamics of financial assets.

3.1 Ito's integral

Definition 3.1. Let $\mathcal{V} = \mathcal{V}(S, T)$ the class of functions

$$f(t, \omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R},$$

such that

- $f(t, \omega)$ is $\mathcal{B} \times \mathcal{F}$ -measurable, where $\mathcal{B}$ is the Borel σ -algebra on $[0, \infty)$ and $\mathcal{F}$ is the σ -algebra on Ω ,
- $f(t, \omega)$ is $\mathcal{F}_t$ adapted.
- $\mathbb{E} \left[\int_S^T f(t, \omega)^2 dt \right] < \infty$.

We will start by defining the Itô integral for *elementary functions*, then this can be extended to functions in $\mathcal{V}$.

Definition 3.2. A function ϕ is called *elementary* if it has the form

$$\phi(t, \omega) = \sum_j e_j(\omega) \mathbb{1}_{[t_j, t_{j+1})}(t).$$

For elementary functions we define the Ito integral as

$$\int_S^T \phi(t, \omega) dW_t(\omega) = \sum_j e_j(\omega) [W_{t_{j+1}} - W_{t_j}](\omega).$$

Lemma 3.1 (Ito isometry). *Let $\phi(t, \omega)$ be a bounded elementary function, then*

$$\mathbb{E} \left[\left(\int_S^T \phi(t, \omega) dW_t(\omega) \right)^2 \right] = \mathbb{E} \left[\int_S^T \phi(t, \omega)^2 dt \right].$$

Using the Ito isometry and the definition of the Ito integral for elementary functions, the Ito integral can be extended to the class of functions $\mathcal{V}$, using that the functions in $\mathcal{V}$ can be expressed as the limit of sequences of simpler functions. It is worth mentioning that the Itô isometry still holds for any function in $\mathcal{V}$.

The result of the integral is a stochastic process, which is a continuous martingale. This is similar to regular integration, where the result of integrating a function is a new function, in this case the result of integrating a process "over" a stochastic process is a new stochastic process, plus the resulting process has the nice property of being a martingale.

The class $\mathcal{V}$ is rather limited and does not even contain all continuous functions, see [21]. Having to restrict to this class would be rather inconvenient when modeling using SDEs, luckily the Itô integral can be extended to a broader class by relaxing the last condition of Definition 3.1

$$\mathbb{P} \left(\int_S^T f(t, \omega)^2 dt < \infty \right) = 1,$$

will call this class of functions $\mathcal{W}$. Notice that the class $\mathcal{W}$ does contain all continuous functions.

3.2 Ito's formula

Having defined the Ito integral, would be interesting to have a way to easily compute stochastic integrals without having to compute the limit of elementary functions, that is the purpose of the Ito's lemma or Ito's formula. The Ito's formula is a stochastic version of the chain rule for derivatives.

Definition 3.3 (Ito process). Let W_t be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. An Ito process is a stochastic process X_t on $(\Omega, \mathcal{F}, \mathbb{P})$ of the form

$$(3.3) \quad X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s,$$

where μ and σ are adapted, measurable processes, that satisfy the integrability conditions

$$\mathbb{P} \left(\int_0^T |\mu(t, \omega)| dt < \infty \right) = 1,$$

and

$$\mathbb{P} \left(\int_0^T \sigma(t, \omega)^2 dt < \infty \right) = 1.$$

The equation (3.3) can be also written in differential form as

$$dX_t = \mu dt + \sigma dW_t.$$

Theorem 3.2 (1-dimensional Itô's formula). Let X_t be an Itô process given by

$$dX_t = \mu dt + \sigma dW_t.$$

Let $g(t, x) \in C^2([0, \infty) \times \mathbb{R})$, then

$$Y_t = g(t, X_t),$$

is an Ito process given by

$$(3.4) \quad dY_t = \frac{\partial g}{\partial t}(t, X_t) dt + \frac{\partial g}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2}(t, X_t) (dX_t)^2,$$

where $(dX_t)^2$ is computed according to the rules

$$(dW_t)^2 = dt, \quad (dt)^2 = 0, \quad dW_t \cdot dt = 0.$$

For the proof of the Ito's formula we refer to [20] Theorem 4.1.2. The formula can be extended to multiple dimensions.

Definition 3.4 (n -dimensional Ito process). Let $W_t = (W_1(t), \dots, W_m(t))$ be a m -dimensional Brownian motion, let μ_i and σ_{ij} be adapted, measurable processes such that

$$\mathbb{P} \left(\int_0^T |\mu_i(t, \omega)| dt < \infty \right) = 1, \quad \mathbb{P} \left(\int_0^T \sigma_{ij}(t, \omega)^2 dt < \infty \right) = 1,$$

then we define the n -dimensional Ito process $X_t = (X_1(t), \dots, X_n(t))$ as

$$\begin{cases} dX_1(t) = \mu_1 dt + \sigma_{11} dW_1(t) + \dots + \sigma_{1m} dW_m(t), \\ \vdots \\ dX_n(t) = \mu_n dt + \sigma_{n1} dW_1(t) + \dots + \sigma_{nm} dW_m(t), \end{cases}$$

or in matrix notation

$$dX_t = \mu dt + \sigma dW_t,$$

where $\mu = (\mu_1, \dots, \mu_n)$ and $\sigma = (\sigma_{ij})$ are an n -dimensional vector and an $n \times m$ matrix respectively.

Theorem 3.3 (General Itô's formula). *Let X_t be an n -dimensional Itô process given by*

$$dX_t = \mu dt + \sigma dW_t.$$

Let $g(t, x) = (g_1(t, x), \dots, g_p(t, x)) \in C^2([0, \infty) \times \mathbb{R}^n)$, then

$$Y_t = g(t, X_t),$$

is a p -dimensional Itô process whose components, $0 \leq k \leq p$, are given by

$$(3.5) \quad dY_k(t) = \frac{\partial g_k}{\partial t}(t, X_t)dt + \sum_{0 \leq i \leq n} \frac{\partial g_k}{\partial x_i}(t, X_t)dX_i(t) + \frac{1}{2} \sum_{0 \leq i, j \leq n} \frac{\partial^2 g_k}{\partial x_i \partial x_j}(t, X_t)dX_i(t)dX_j(t),$$

where $dX_i(t)dX_j(t)$ is computed according to the rules

$$(dW_i(t))^2 = dt, \quad dW_i(t)dW_j(t) = 0 \quad (i \neq j).$$

3.3 Stochastic differential equations

A **stochastic differential equation** (SDE) is an equation of the form

$$(3.6) \quad dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X_0 = x_0,$$

where $b : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the *drift* and $\sigma : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is the *diffusion* coefficient. A solution is an Itô process X_t satisfying (3.6) almost surely.

The following theorem, whose proof may be found in [20, Thm. 5.2.1], gives sufficient conditions for a solution to exist and be unique.

Theorem 3.4 (Existence and uniqueness). *Let $T > 0$. Suppose b and σ satisfy, for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,*

1. (Lipschitz) $|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq C|x - y|$,
2. (Linear growth) $|b(t, x)|^2 + |\sigma(t, x)|^2 \leq C^2(1 + |x|^2)$,

for some constant $C > 0$, and let $x_0 \in \mathbb{R}^n$. Then the SDE (3.6) has a unique (up to indistinguishability) continuous adapted solution $\{X_t\}_{0 \leq t \leq T}$.

We close with two fundamental examples, both following [20, Ch. 5].

3.3.1 Example: Geometric Brownian motion

Let $\mu, \sigma \in \mathbb{R}$ with $\sigma \neq 0$ and $S_0 > 0$. Consider the SDE

$$(3.7) \quad dS_t = \mu S_t dt + \sigma S_t dW_t.$$

This is called **geometric Brownian motion** and is the model used in Section 4.2 for the dynamics of a stock price.

Applying Theorem 3.2 to $g(t, x) = \ln x$, with $g_t = 0$, $g_x = x^{-1}$, $g_{xx} = -x^{-2}$,

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 = \mu dt + \sigma dW_t - \frac{\sigma^2}{2} dt = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t.$$

Integrating both sides from 0 to t gives $\ln(S_t/S_0) = (\mu - \sigma^2/2)t + \sigma W_t$, so the unique solution is

$$(3.8) \quad S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right].$$

3.3.2 Example: Ornstein-Uhlenbeck process

Let $\theta, \sigma > 0$ and $\mu \in \mathbb{R}$. Consider the SDE

$$(3.9) \quad dX_t = \theta(\mu - X_t) dt + \sigma dW_t, \quad X_0 = x_0.$$

This is the **Ornstein-Uhlenbeck (OU) process**, a mean-reverting model: the drift $\theta(\mu - X_t)$ pulls the process toward the long-run mean μ at rate θ . It appears, for instance, in the Vasicek interest rate model.

To solve, let $Y_t = (X_t - \mu)e^{\theta t}$. Applying Theorem 3.2 with $g(t, x) = (x - \mu)e^{\theta t}$,

$$dY_t = \theta(X_t - \mu)e^{\theta t} dt + e^{\theta t} dX_t = e^{\theta t} [\theta(\mu - X_t) + \theta(X_t - \mu)] dt + \sigma e^{\theta t} dW_t = \sigma e^{\theta t} dW_t.$$

Integrating and reverting the substitution,

$$(3.10) \quad X_t = \mu + (x_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s.$$

The OU process is Gaussian with $\mathbb{E}[X_t] = \mu + (x_0 - \mu)e^{-\theta t} \rightarrow \mu$ as $t \rightarrow \infty$.

BLACK-SCHOLES

Black-Scholes is a mathematical model for pricing options, originally published by Black-Scholes [5] and Merton [18]. Although many other models have been developed since, the Black-Scholes model remains as the most important model in option pricing, as in many cases options in markets are quoted using the Black-Scholes formula.

A derivative is a financial instrument whose payments are derived from the price of an other asset, called the underlying asset. Derivatives are widely used in finance, whether to hedge financial risk, this is to protect against adverse price movements, or to bet on specific views, especially when it's difficult to take a position in the underlying asset. So being able to price them and justify the price is key for the market to function.

4.1 Replication and arbitrage

One of the most important concepts in modern finance is the so called *law of one price*. The idea is that if two financial assets have exactly the same payoffs in all possible scenarios, then they should have the same price.

This is linked to the concept of *arbitrage*, informally we say that arbitrage is the opportunity to make a riskless profit. For example, if two assets that have exactly the same payoffs in all possible scenarios, but don't follow the law of one price, one can buy the cheaper one and sell the more expensive one, gaining an instant profit and compensating the payoffs of the sold asset using the payoffs of the bought one, which match exactly.

We will work under the assumption that we are operating in a market with no arbitrage, this assumption is technically known as *The First Fundamental Theorem of Asset Pricing*, and a key part of the *Efficient Market Hypothesis*. Under this assumption, we can price any derivative

by creating a replicating portfolio, this is a set of assets that has exactly the same payoffs as the derivative in all possible scenarios.

There are two main ways of creating a replicating portfolio:

- **Static replication** refers to an replicating portfolio that is created at the beginning and held as is until maturity.
- **Dynamic replication** refers to a replicating portfolio that requires a continuous change of its composition over time.

For a further discussion on replication, and its applications in finance, we recommend the book [9].

We will show two examples of static replication from [21] that will be useful to understand the Black-Scholes model.

4.1.1 Example: Forward contract

Assume that there is an agent that borrows and lends money at a continuously compounded rate r , known as the **risk-free rate**. This means that if we borrow 1\$ at time $t = 0$, we will have to pay e^{rT} \$ at time $t = T$.

A forward contract gives the holder the obligation to buy an asset at a fixed price K at a fixed date T .

To price this contract imagine we are at time t and the current price of an asset is S , we want to buy a forward contract with maturity T and fixed price K .

We have two options, we can buy the contract at time t for a price of F and at time T pay K to get the asset. Alternatively we can buy the asset for S and borrow $e^{-r(T-t)}K$ at time t , so at time T we will pay $K = e^{r(T-t)}e^{-r(T-t)}K$ while having the asset. Both options have the same cashflow at time T , pay K and own the asset, so either we have

$$F = S - e^{-r(T-t)}K,$$

or there is an arbitrage opportunity.

We have found the price of the forward contract by replicating its cashflows, using the underlying asset and the risk-free rate, plus arguing that there should not be an arbitrage opportunity.

4.1.2 Example: Put-Call parity

Definition 4.1 (European call option). An **European call option** is a financial contract that gives the holder the right, but not the obligation, to **buy** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European call option at maturity T is given by:

$$(4.1) \quad C_T = \max(S_T - K, 0),$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

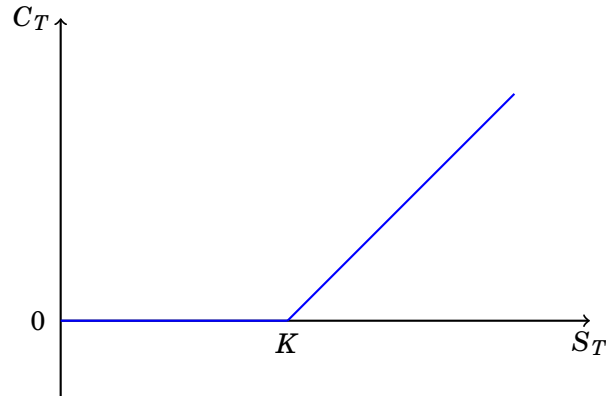


Figure 4.1: Payoff of a European call option

Definition 4.2 (European put option). An **European put option** is a financial contract that gives the holder the right, but not the obligation, to **sell** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European put option at maturity T is given by:

$$(4.2) \quad P_T = \max(K - S_T, 0),$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

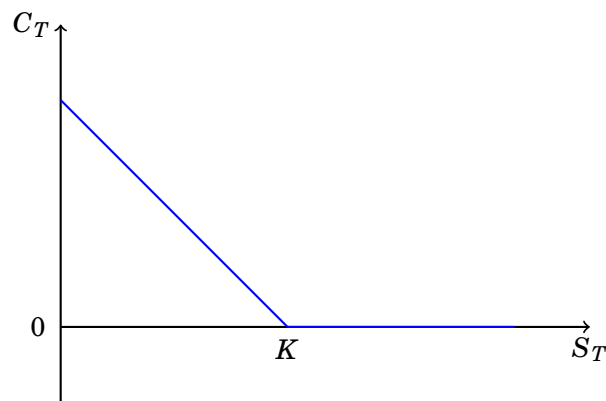


Figure 4.2: Payoff of a European put option

Now consider a portfolio where we buy a call option and sell a put option with same strike K and maturity T , the payoff of such portfolio at time T will be exactly $S_T - K$. So we are essentially

receiving the asset and paying K at time T , as we have seen this is the payoff of a forward contract, which means that at time t the price of our portfolio must be equal to the price of a forward of the asset,

$$(4.3) \quad C_t - P_t = F_t = S_t - e^{-r(T-t)}K,$$

this formula is called *put-call parity* and illustrates the relationship between the price of a call and a put option.

4.2 The Black-Scholes model

The Black-Scholes model is a mathematical model for pricing European options, calls or puts, that is based on replicating the payoffs of the option dynamically using the underlying asset and a risk-free rate bond.

Like before we assume that the market is arbitrage-free, and that there is a constant risk-free rate r . As the model is based on dynamic replication, we also need to assume that we can rebalance our replicating portfolio continuously, i.e. we can buy and sell any amount of the underlying asset and the risk-free rate bond instantaneously at any time.

Let S_t be the price of the underlying asset at time t and β_t be the price of the risk-free rate bond at time t . The dynamics of this two assets are given by the following stochastic differential equations (SDEs):

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad d\beta_t = r\beta_t dt.$$

To construct the replicating portfolio. Let a_t be the amount of the underlying asset in our replicating portfolio and b_t be the amount of the risk-free rate bond in our replicating portfolio. Then the value of the portfolio at any time t is given by:

$$V_t = a_t S_t + b_t \beta_t.$$

We need this portfolio to have the same payoff as the call option at time T , so

$$V_T = \max(S_T - K, 0).$$

As the portfolio is dynamically replicating the option, the cashflow of the portfolio must match the ones of the option, these are the payoff as seen before, and the initial payment, so no cash must come in or out of the portfolio at any time between t and T . This means that the portfolio must be *self-financing*, so the rebalanced must come from selling the underlying asset and buying the bond, or vice versa. This is expressed mathematically as

$$(4.4) \quad dV_t = a_t dS_t + b_t d\beta_t,$$

which can be extended to

$$dV_t = (a_t \mu S_t + b_t r \beta_t) dt + a_t \sigma S_t dW_t.$$

Now as the value of the replicating portfolio V_t should be equal to the value of the option at any time t , we assume that this can be expressed by a function $V_t = f(t, S_t)$, depending only on the time t and the price of the underlying asset S_t . Thus applying Ito's formula we have that

$$dV_t = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} \mu S_t + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right) dt + \frac{\partial f}{\partial S} \sigma S_t dW_t,$$

matching the two expressions for dV_t we have that

$$a_t = \frac{\partial f}{\partial S},$$

and

$$b_t = \frac{1}{r\beta_t} \left(\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right)$$

as V_t is the value of the replicating portfolio and the option at time t , we have that

$$f(t, S_t) = V_t = a_t S_t + b_t \beta_t,$$

using the values of a_t and b_t from above,

$$f(t, S_t) = \frac{\partial f}{\partial S} S_t + \frac{1}{r} \left(\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S_t^2 \right),$$

that can be rearranged to

$$(4.5) \quad \frac{\partial f}{\partial t} = r f(t, S) - r \frac{\partial f}{\partial S} S - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2,$$

with terminal condition

$$f(T, S) = \max(S - K, 0),$$

which is the **Black-Scholes PDE**.

Note that we dropped the subscript in S_t . The relation between S and t was at the trajectories described through it's dynamics, but at time t the underlying can take any positive value, $S \geq 0$.

4.3 Pricing - Solving the Black-Scholes PDE

There is an analytical solution to the Black-Scholes PDE. Our derivation follows the approach of [21, Ch. 11.4], adapted to **log-moneyness** coordinates $x = \ln(S/K)$ instead of the log-price $\ln S$ used there; the two differ only by the constant shift $\ln K$ and the subsequent reduction to the heat equation is structurally identical.

Start by change of variables. Let $\tau = T - t$ be the **time to maturity**, and $x = \ln\left(\frac{S}{K}\right)$ be the log-moneyness. We define a new function $u(\tau, x) = f(t, S)$ so the Black-Scholes PDE is transformed to,

$$\frac{\partial u}{\partial \tau} = \frac{1}{2} \sigma^2 \frac{\partial^2 u}{\partial x^2} + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\partial u}{\partial x} - r u(\tau, x).$$

Now we would like to remove the drift term (the derivative with respect to x) and the zero order term, so we have a standard heat equation. To do it we do another transformation,

$$u(x, \tau) = e^{\alpha x + \beta \tau} v(x, \tau),$$

where α and β are the coefficients that lead to the heat equation, using,

$$\begin{aligned} u_\tau &= e^{\alpha x + \beta \tau} (\beta v + v_\tau), \\ u_x &= e^{\alpha x + \beta \tau} (\alpha v + v_x), \\ u_{xx} &= e^{\alpha x + \beta \tau} (\alpha^2 v + 2\alpha v_x + v_{xx}), \end{aligned}$$

we get,

$$v_\tau = \frac{1}{2} \sigma^2 v_{xx} + v_x \left(\sigma^2 \alpha + r - \frac{1}{2} \sigma^2 \right) + v \left(\frac{1}{2} \sigma^2 (\alpha^2 - \alpha) + r\alpha - r - \beta \right),$$

thus,

$$\alpha = \frac{1}{2} - \frac{r}{\sigma^2} = -\frac{1}{\sigma^2} \left(r - \frac{1}{2} \sigma^2 \right),$$

and

$$\begin{aligned} \beta &= \alpha \left(r + \frac{1}{2} \sigma^2 (\alpha - 1) \right) - r \\ &= \alpha \left(\frac{1}{2} r - \frac{1}{4} \sigma^2 \right) - r \\ &= -\frac{1}{2\sigma^2} \left(r - \frac{1}{2} \sigma^2 \right)^2 - r \\ &= -\frac{1}{2\sigma^2} \left(r + \frac{1}{2} \sigma^2 \right)^2. \end{aligned}$$

So we get the standard heat equation,

$$\frac{\partial v}{\partial \tau} = \frac{1}{2} \sigma^2 \frac{\partial^2 v}{\partial x^2},$$

with initial conditions for v ,

$$v(x, 0) = e^{-\alpha x} u(x, 0) = K e^{-\alpha x} \max(e^x - 1, 0).$$

Now we can solve the equation using the heat operator,

$$v(x, \tau) = \int_{-\infty}^{\infty} v(y, 0) k(\sigma^2; \tau, x - y) dy,$$

where the function k is the *heat kernel*, i.e. the fundamental solution of the heat equation $v_\tau = \lambda v_{xx}$, given by

$$(4.6) \quad k(\lambda; t, x) := \frac{1}{\sqrt{4\lambda\pi t}} \exp\left(-\frac{x^2}{4\lambda t}\right),$$

see [21, Ch. 11.4]. Using the initial condition we get,

$$v(x, \tau) = \int_0^\infty K e^{-\alpha y} (e^y - 1) k\left(\frac{\sigma^2}{2}; \tau, x - y\right) dy,$$

this leaves us with two integrals

$$I_1 = K \int_0^\infty e^{(1-\alpha)y} k\left(\frac{\sigma^2}{2}; \tau, x-y\right) dy,$$

$$I_2 = -K \int_0^\infty e^{-\alpha y} k\left(\frac{\sigma^2}{2}; \tau, x-y\right) dy,$$

if we make a change of variable, $s = x - y$, then

$$I_1 = K e^{(1-\alpha)x} \int_{-\infty}^x e^{-(1-\alpha)s} k\left(\frac{\sigma^2}{2}; \tau, s\right) ds,$$

$$I_2 = -K e^{-\alpha x} \int_{-\infty}^x e^{\alpha s} k\left(\frac{\sigma^2}{2}; \tau, s\right) ds.$$

To evaluate both integrals we use the following identity, whose proof is a standard completion of the square.

Lemma 4.1. *Let Φ denote the standard Gaussian cumulative distribution function,*

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

Then for any $a, b \in \mathbb{R}$ and $\sigma, \tau > 0$,

$$(4.7) \quad \int_{-\infty}^a k\left(\frac{\sigma^2}{2}; \tau, s\right) e^{-bs} ds = e^{\frac{b^2 \sigma^2 \tau}{2}} \Phi\left(\frac{a}{\sigma \sqrt{\tau}} + b \sigma \sqrt{\tau}\right).$$

Proof. Completing the square in the exponent of the integrand,

$$-\frac{s^2}{2\sigma^2\tau} - bs = -\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau} + \frac{b^2\sigma^2\tau}{2},$$

so pulling out the constant factor and applying the change of variable $u = (s + b\sigma^2\tau)/(\sigma\sqrt{\tau})$,

$$\begin{aligned} \int_{-\infty}^a k\left(\frac{\sigma^2}{2}; \tau, s\right) e^{-bs} ds &= e^{\frac{b^2 \sigma^2 \tau}{2}} \int_{-\infty}^a \frac{1}{\sqrt{2\pi\sigma^2\tau}} \exp\left(-\frac{(s + b\sigma^2\tau)^2}{2\sigma^2\tau}\right) ds \\ &= e^{\frac{b^2 \sigma^2 \tau}{2}} \int_{-\infty}^{\frac{a + b\sigma^2\tau}{\sigma\sqrt{\tau}}} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = e^{\frac{b^2 \sigma^2 \tau}{2}} \Phi\left(\frac{a}{\sigma\sqrt{\tau}} + b\sigma\sqrt{\tau}\right). \end{aligned}$$

□

Applying Lemma 4.1 to I_1 (with $a = x$, $b = 1 - \alpha$) and to I_2 (with $a = x$, $b = -\alpha$), and substituting back to $u = e^{\alpha x + \beta \tau} v$, we get

$$I_1 \Rightarrow I_3 = K \exp\left(\alpha x + \beta \tau + (1 - \alpha)x + \frac{(1 - \alpha)^2 \sigma^2 \tau}{2}\right) \Phi(d_1),$$

$$I_2 \Rightarrow I_4 = -K \exp\left(\alpha x + \beta \tau - \alpha x + \frac{\alpha^2 \sigma^2 \tau}{2}\right) \Phi(d_2),$$

where

$$d_1 = \frac{x}{\sigma\sqrt{\tau}} + (1 - \alpha)\sigma\sqrt{\tau},$$

$$d_2 = \frac{x}{\sigma\sqrt{\tau}} - \alpha\sigma\sqrt{\tau} = d_1 - \sigma\sqrt{\tau},$$

rearranging and simplifying the expressions

$$I_3 = K \exp\left(x + \left(\beta + \frac{(1-\alpha)^2\sigma^2}{2}\right)\tau\right) \Phi(d_1),$$

$$I_4 = -K \exp\left(\left(\beta + \frac{\alpha^2\sigma^2}{2}\right)\tau\right) \Phi(d_2),$$

the exponent of the first term simplifies to

$$x + \left(\beta + \frac{(1-\alpha)^2\sigma^2}{2}\right)\tau = x + \frac{1}{2\sigma^2} \left(-\left(r + \frac{1}{2}\sigma^2\right)^2 + \left(r + \frac{1}{2}\sigma^2\right)^2\right)\tau = x,$$

and for the second one we use the fact that while deriving β , on the second equality we found

$$\beta = \frac{1}{2}\alpha \left(r - \frac{1}{2}\sigma^2\right) - r$$

multiplying by $1 = \frac{\sigma^2}{\sigma^2}$ and rearranging we get

$$\beta = \frac{\sigma^2}{2}\alpha \frac{1}{\sigma^2} \left(r - \frac{1}{2}\sigma^2\right) - r$$

and using the definition of α we get

$$\beta = -\frac{\sigma^2}{2}\alpha^2 - r.$$

Thus

$$I_3 = K e^x \Phi(d_1),$$

$$I_4 = -K e^{-r\tau} \Phi(d_2).$$

Finally, reverting the substitutions $\tau = T - t$ and $x = \ln(S/K)$ we get the **Black-Scholes formula**

$$(4.8) \quad f(t, S) = S \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

where

$$\begin{aligned} d_1 &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + (1-\alpha)\sigma\sqrt{T-t} \\ &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + \left(1 + \frac{1}{\sigma^2} \left(r - \frac{1}{2}\sigma^2\right)\right) \sigma\sqrt{T-t} \\ &= \frac{\ln\left(\frac{S}{K}\right)}{\sigma\sqrt{T-t}} + \left(\frac{1}{\sigma^2} \left(\sigma^2 + r - \frac{1}{2}\sigma^2\right)\right) \sigma\sqrt{T-t} \\ &= \frac{1}{\sigma\sqrt{T-t}} \left(\ln\left(\frac{S}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right), \end{aligned}$$

and, of course

$$\begin{aligned} d_2 &= d_1 - \sigma\sqrt{T-t} \\ &= \frac{1}{\sigma\sqrt{T-t}} \left(\ln\left(\frac{S}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right). \end{aligned}$$

4.4 Calibration - Implied volatility

The Black-Scholes formula (4.8) gives the call price as a function of market observables (t, S, r) , contract parameters (K, T) , and the single volatility parameter σ . In practice, market prices $C^{\text{mkt}}(K, T)$ are observable for a range of strikes and maturities, and the question is: which value of σ is implied by each observed price?

Definition 4.3 (Implied volatility). Given a market price C^{mkt} for a European call with strike K and maturity T , the **implied volatility** $\hat{\sigma}(K, T)$ is the unique $\sigma > 0$ such that

$$(4.9) \quad C_{\text{BS}}(t, S; K, T, \hat{\sigma}) = C^{\text{mkt}},$$

where C_{BS} denotes the Black-Scholes formula (4.8).

Uniqueness follows from strict monotonicity of the call price in σ . To see this we compute the **vega** of the option, $\mathcal{V} := \partial C_{\text{BS}} / \partial \sigma$.

Write $\tau = T - t$ for brevity and note that d_1 and d_2 both depend on σ . Splitting d_1 into its two terms makes differentiation straightforward,

$$d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau},$$

so

$$(4.10) \quad \frac{\partial d_1}{\partial \sigma} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} + \frac{\sqrt{\tau}}{2}.$$

Since $d_2 = d_1 - \sigma\sqrt{\tau}$,

$$(4.11) \quad \frac{\partial d_2}{\partial \sigma} = \frac{\partial d_1}{\partial \sigma} - \sqrt{\tau} = -\frac{\ln(S/K) + r\tau}{\sigma^2\sqrt{\tau}} - \frac{\sqrt{\tau}}{2}.$$

In particular,

$$(4.12) \quad \frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma} = \sqrt{\tau}.$$

Differentiating (4.8) with the chain rule gives

$$\frac{\partial C_{\text{BS}}}{\partial \sigma} = S\phi(d_1)\frac{\partial d_1}{\partial \sigma} - Ke^{-r\tau}\phi(d_2)\frac{\partial d_2}{\partial \sigma},$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ is the standard Gaussian density. The two terms simplify if we use the identity

$$(4.13) \quad S\phi(d_1) = Ke^{-r\tau}\phi(d_2).$$

To verify it, note that $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = 2(\ln(S/K) + r\tau)/(\sigma\sqrt{\tau})$, so

$$d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = 2\ln(S/K) + 2r\tau.$$

Therefore, since the $1/\sqrt{2\pi}$ factors cancel in the ratio,

$$\frac{\phi(d_2)}{\phi(d_1)} = \frac{e^{-d_2^2/2}}{e^{-d_1^2/2}} = e^{(d_1^2 - d_2^2)/2} = \exp\left(\ln \frac{S}{K} + r\tau\right) = \frac{S}{K} e^{r\tau},$$

which is exactly (4.13). Applying it and using (4.12),

$$(4.14) \quad \mathcal{V} := \frac{\partial C_{BS}}{\partial \sigma} = S \phi(d_1) \left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma} \right) = S \sqrt{T-t} \phi(d_1) > 0.$$

Since $\mathcal{V} > 0$, the map $\sigma \mapsto C_{BS}(\sigma)$ is a strictly increasing bijection from $(0, \infty)$ onto the no-arbitrage interval $(\max(S - Ke^{-r(T-t)}, 0), S)$, so the implied volatility exists and is unique for any market price in this range.

Since (4.9) has no closed-form solution in σ , the implied volatility is computed numerically. The smoothness and strict positivity of $\mathcal{V}$ make Newton's method an efficient choice: starting from an initial guess $\sigma^{(0)} > 0$, one iterates

$$(4.15) \quad \sigma^{(n+1)} = \sigma^{(n)} - \frac{C_{BS}(\sigma^{(n)}) - C^{\text{mkt}}}{\mathcal{V}(\sigma^{(n)})},$$

until $|C_{BS}(\sigma^{(n)}) - C^{\text{mkt}}| < \varepsilon$.

4.4.1 The implied volatility surface

If Black-Scholes with a single σ perfectly described the market, every option on the same underlying would yield the same implied volatility. In practice, the map $(K, T) \mapsto \hat{\sigma}(K, T)$ is non-trivial, and the resulting object is called the **implied volatility surface**.

Two features of this surface are well documented in the literature, see [9]:

- **Volatility skew:** for a fixed maturity T , equity index options exhibit a pronounced *negative skew*: implied volatility increases as K decreases relative to S . This reflects the demand for downside protection and the asymmetry of return distributions.
- **Term structure:** for a fixed K , implied volatility varies with maturity, reflecting the market's expectation of future realized volatility at different horizons.

The non-flatness of the surface is direct empirical evidence that constant- σ Black-Scholes is inconsistent with observed market prices. This motivates more flexible models such as the *local volatility model*, which replaces σ by a deterministic function $\sigma(t, S)$ calibrated to reproduce the full surface exactly, as described in the following section.

Figure 4.3 shows the implied volatility surface computed from SPY option quotes on 2026-04-30, using the Black-Scholes-Merton formula with a continuously compounded dividend yield calibrated per maturity from put-call parity (see Appendix A for implementation details). The characteristic features are clearly visible: the volatility skew in the moneyness direction and the term structure in the maturity direction. Figure 4.4 shows the smile slice at the median maturity

($T \approx 0.21$ yr), where **out of the money**, OTM¹ puts (red) and OTM calls (blue) meet smoothly at the money.

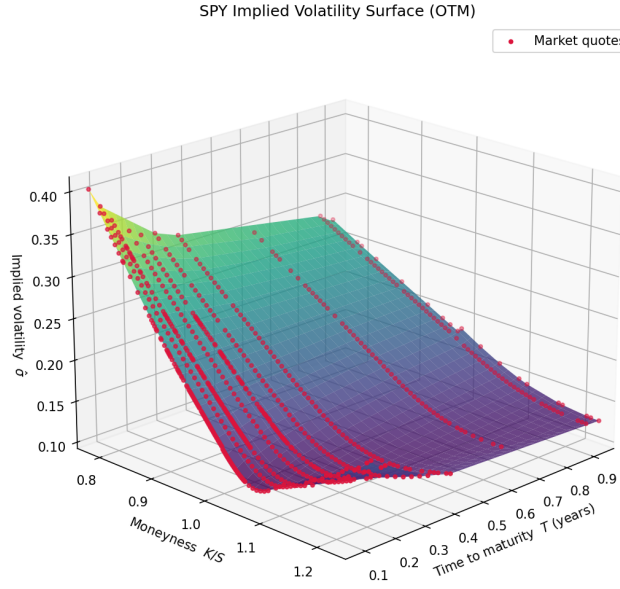


Figure 4.3: SPY implied volatility surface (OTM composite) on 2026-04-30. Axes: time to maturity T (years), moneyness K/S (inverted scale), implied volatility $\hat{\sigma}$. Red dots are individual market quotes; the surface is linearly interpolated.

¹A call option is said to be out of the money, OTM, if the strike is greater than the price, $K > S$, this is because the price of the underlying have to move in the direction of the strike K to make the option worth something at expiration. Similarly a call option is said to be at the money, ATM, if $K = S$ and in the money, ITM, if $K < S$. Is the reverse for put options, a put option is OTM if $K < S$ and ITM if $K > S$.

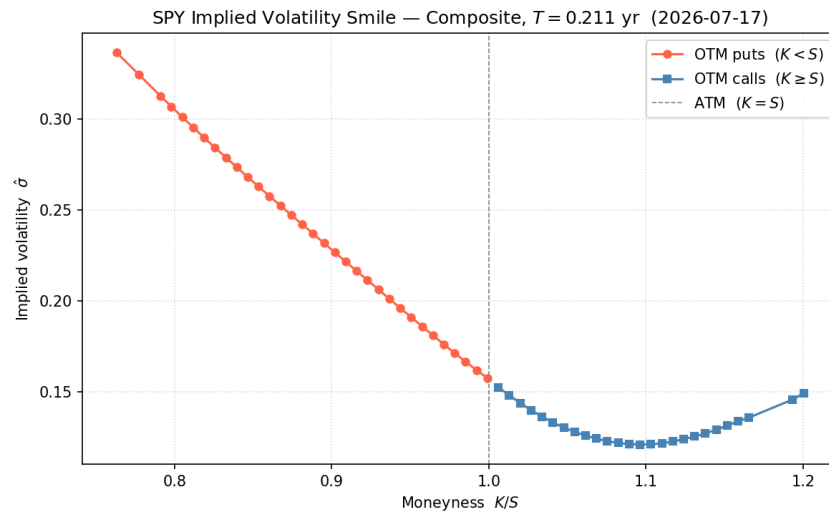


Figure 4.4: Volatility smile at the median maturity ($T \approx 0.21$ yr, expiration 2026-07-17). OTM puts (red) and OTM calls (blue) are joined from the calibrated composite surface. The smooth join at $K = S$ reflects the consistent dividend yield calibration.

ARBITRAGE THEORY

We can generalize the approach used to derive the Black-Scholes PDE for a more general products. Thus in this section will try to give a general framework to derive equations to price a **contingent claim**. A contingent claim is a financial instrument whose payoff depends on the outcome of uncertain future events, for example an European option.

Throughout the chapter we work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$, where $\{\mathcal{F}_t\}$ is the filtration generated by a k -dimensional Brownian motion $W(t)$, so $\mathcal{F}_t$ is the information available from the market up to time t .

Definition 5.1 (Market). A market is a set of $n + 1$ assets $S_0, S_1, \dots, S_n$, where the price of the riskless S_0 is given by

$$\frac{dS_0(t)}{S_0} = r(S_0(t), t)dt,$$

and the prices of the other risky assets are given by

$$dS_i(t) = \mu_i(S(t), t)dt + \sigma_i(S(t), t)^\top dW(t), \quad i = 1, \dots, n,$$

where μ_i is real valued and σ_i takes values in $\mathbb{R}^k$.

Definition 5.2 (Portfolio). A portfolio in the market $\{S_i(t)\}$ is an $n + 1$ dimensional $\mathcal{F}_t$ -adapted stochastic process

$$\theta(t) = (\theta_0(t), \theta_1(t), \dots, \theta_n(t)),$$

and its value at time t is given by

$$V(t) = \theta(t)^\top S(t) = \sum_{i=0}^n \theta_i(t) S_i(t).$$

Note that as a portfolio $\theta(t)$ is an adapted process it doesn't use future information, so it is a generalization on what in reality would be to rebalance ones investments in the market at time t .

We can generalize the self-financing condition presented in (4.4) as,

$$(5.1) \quad V(t) = V(0) + \int_0^t \theta(u)^\top dS(u),$$

where the integral is a vector of stochastic integrals taken component wise,

$$\int_0^t \theta(u)^\top dS(u) = \sum_{i=0}^n \int_0^t \theta_i(u) dS_i(u),$$

in differential form

$$(5.2) \quad dV(t) = \theta(t)^\top dS(t).$$

As our market consists of assets that are described as stochastic processes, an European contingent claim paying at time T is a function $f(S(T), T)$ depending on these assets, which is a random variable as $S(T)$ is a random vector of the values of the assets at time T .

5.1 PDE method

Consider an European contingent claim with payoff $f(S(T))$, similar to what we did in section 4.2, we want to construct a replicating portfolio $V(S(t), t)$ that has the same payoff as the contingent claim at time T . We follow [15, Ch. 1], see also [4, Ch. 13] for a detailed treatment. Thus using the Ito formula (3.5) we have

$$\begin{aligned} dV(S(t), t) &= \frac{\partial V}{\partial t} dt + \sum_{0 \leq i \leq n} \frac{\partial V}{\partial S_i} dS_i(t) + \frac{1}{2} \sum_{0 \leq i, j \leq n} \frac{\partial^2 V}{\partial S_i \partial S_j} dS_i(t) dS_j(t) \\ &= \sum_{0 \leq i \leq n} \frac{\partial V}{\partial S_i} dS_i(t) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{0 \leq i, j \leq n} \frac{\partial^2 V}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\top \sigma_j(S(t), t) \right) dt, \end{aligned}$$

where $\sigma_0 \equiv 0$ as S_0 is riskless. Imposing the self-financing condition (5.2) we find that

$$(5.3) \quad \theta_i(t) = \frac{\partial V}{\partial S_i}, \quad \forall i = 0, \dots, n,$$

and

$$(5.4) \quad \frac{\partial V}{\partial t} + \frac{1}{2} \sum_{0 \leq i, j \leq n} \frac{\partial^2 V}{\partial S_i \partial S_j} \sigma_i(S(t), t)^\top \sigma_j(S(t), t) = 0.$$

Also using $V(S(t), t) = \theta(t)^\top S(t)$ in (5.3) we get

$$(5.5) \quad V(S(t), t) = \sum_{i=0}^n \frac{\partial V}{\partial S_i} S_i.$$

The equations (5.4) and (5.5) describe the PDE for the price of the contingent claim, with independent variables $(S, t) \in [0, \infty)^{n+1} \times [0, T]$ and terminal condition the payoff at time T

$$V(S, T) = f(S).$$

As in Black-Scholes model, μ does not explicitly appear in the PDE, its only present implicitly in the prices $S(t)$ of the assets.

To see that this is indeed a generalization of chapter 4, take $n = 1$ with a constant rate r , so that $S_0(t) = e^{rt}$, and $\sigma_1(S, t) = \sigma S_1$. As S_0 is deterministic we can write the price as a function of time and the risky asset alone, $f(t, S) = V(S_0(t), S, t)$, and the chain rule gives

$$\frac{\partial f}{\partial t} = \frac{\partial V}{\partial t} + rS_0 \frac{\partial V}{\partial S_0}.$$

Now (5.4) gives the first term and (5.5) the second,

$$\frac{\partial V}{\partial t} = -\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2}, \quad S_0 \frac{\partial V}{\partial S_0} = f(t, S) - S \frac{\partial f}{\partial S},$$

and substituting we recover the Black-Scholes PDE (4.5). Note that the rate r enters the pricing equation only through (5.5), i.e. through the position held in the riskless asset.

5.2 Martingale method

The martingale method is an alternative framework to the PDE method, that allows to price contingent claims in a more flexible way.

We will use [4, Ch. 10] as a reference for this section, we start with a couple of definitions related with the concept of market and portfolio defined previously.

Definition 5.3 (Attainable process). We say a price process $V(t)$ is **attainable** if there exists a self-financing portfolio $\theta(t)$ such that $V(t) = \theta(t)^\top S(t)$.

The attainable process is a formalization of the dynamic replicating portfolio idea. An other restriction on the portfolios we are going to consider is the following,

Definition 5.4 (Admissible portfolio). A portfolio $\theta(t)$ is called **admissible** if exists $\alpha > 0$ such that

$$\int_0^t \theta(u)^\top dS(u) \geq -\alpha, \quad \forall t \in [0, T].$$

In real life we can interpret this condition as a limit on the amount the portfolio can lose or borrow, as bank, clearing house or counterparty will not allow such risks and would force to cancel the deal. Technically also allow to cancel out some other undesirable strategies that would prevent some key theorems to hold, see [21, Sec. 14.5 and 14.6] for more details on such strategies.

Now we will formalize the idea of arbitrage in a market.

Definition 5.5 (Arbitrage). A self-financing and admissible portfolio $\theta(t)$ is an **arbitrage** if one of the following conditions hold for time t :

1. $\theta(0)^\top S(0) < 0$ and $\mathbb{P}(\theta(t)^\top S(t) \geq 0) = 1$,
2. $\theta(0)^\top S(0) = 0$ and $\mathbb{P}(\theta(t)^\top S(t) \geq 0) = 1$ and $\mathbb{P}(\theta(t)^\top S(t) > 0) > 0$,

Consider that the **risk-free** asset S_0 has a constant price of 1, i.e. the risk-free rate is $r(S(t), t) \equiv 0$. Under this assumption we introduce the **risk-neutral measure** $\mathbb{Q}$ as,

Definition 5.6. A probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ is called a **risk-neutral measure** (also known as **equivalent martingale measure**) if

- $\mathbb{Q}$ is equivalent to $\mathbb{P}$ on $\mathcal{F}_T$, that is, both measures have the same null sets,

$$\mathbb{Q}(A) = 0 \iff \mathbb{P}(A) = 0, \quad \forall A \in \mathcal{F}_T,$$

so they agree on which events are impossible, but not on how likely the possible ones are.

- All price process in the market $S_0, S_1, \dots, S_n$ are martingales under $\mathbb{Q}$ on the time interval $[0, T]$,

$$\mathbb{E}^{\mathbb{Q}}[S_i(t) | \mathcal{F}_s] = S_i(s), \quad \forall 0 \leq s \leq t \leq T,$$

where $\mathbb{E}^{\mathbb{Q}}$ denotes the expectation taken under $\mathbb{Q}$.

We refer to $\mathbb{P}$ as the "real" probability measure and reflects the true probability of the price processes, whereas $\mathbb{Q}$ is a probability measure under which the expected price of any asset is equal to the current one regardless of their risk (volatility).

Under this assumption we have the following theorem.

Theorem 5.1 (First fundamental theorem of asset pricing). *The market $\{S_i(t)\}$ is arbitrage-free if and only if there exists an equivalent (local) martingale measure $\mathbb{Q}$.*

We include the local martingales¹ as restricting to the martingales would eliminate too many processes, that are of interest.

The restriction on the risk-free asset being constant can be dropped by considering the first price process a numeraire and normalizing the market. For S_0 to be a numeraire we need that $S_0(t) > 0$ almost surely for all $t \geq 0$. Using S_0 as numeraire we can consider the processes $Z_i = S_i(t)/S_0(t)$, which are the **discounted** price processes of the assets.

Definition 5.7 (Normalized market). A normalized market is a vector of processes Z given by

$$Z_i(t) = \frac{S_i(t)}{S_0(t)}, \quad i = 1, \dots, n.$$

¹A process X is a **local martingale** if there is an increasing sequence of stopping times $\tau_m \rightarrow \infty$ such that every stopped process $X_{t \wedge \tau_m}$ is a martingale. Every martingale is a local martingale, but not the other way around, as the integrability condition may fail globally. See [21, Ch. 7].

The results presented so far do not depend on whether we work with the market $\{S_i(t)\}$ as defined in 5.1 or with the normalized market $\{Z_i(t)\}$ of Definition 5.7. A portfolio is self-financing for one market if and only if it is self-financing for the other, and the same holds for being an arbitrage, see [4, Ch. 10]. Theorem 5.1 therefore applies to any market admitting a numeraire, after normalizing it.

Notice that if the numeraire is the risk-free asset S_0 then the market $\{S_i(t)\}$ is arbitrage-free if all assets $S_i(t)$ have a local rate of return $r(t)$ under $\mathbb{Q}$, that is, if $\mu_i(S(t), t) = r(t)S_i(t)$ for all $i = 1, \dots, n$ under $\mathbb{Q}$. This is the sense in which $\mathbb{Q}$ is risk-neutral, under it every asset drifts at the risk-free rate no matter its volatility.

Under the assumption of a market with no arbitrage, a key question is whether a contingent claim can be replicated by a portfolio consisting of the market assets. This yields the following theorem.

Theorem 5.2 (Second fundamental theorem of asset pricing). *Assuming absence of arbitrage, the market model is complete if and only if the martingale measure $\mathbb{Q}$ is unique.*

Finally we can give a general formula to price contingent claims. Under the assumption of a complete market, the price at time t of a contingent claim with payoff $f(S(T))$ is given by

$$(5.6) \quad V(t, f(S(T))) = S_0(t) \mathbb{E}^{\mathbb{Q}} \left[\frac{f(S(T))}{S_0(T)} \middle| \mathcal{F}_t \right],$$

see [4, Ch. 10]. This is just the martingale property of Definition 5.6 read backwards, the normalized price $V(t)/S_0(t)$ is a $\mathbb{Q}$ -martingale whose terminal value is $f(S(T))/S_0(T)$. If the numeraire is the risk-free asset, $S_0(t) = e^{\int_0^t r(s)ds}$ and (5.6) becomes

$$V(t, f(S(T))) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} f(S(T)) \middle| \mathcal{F}_t \right].$$

LOCAL VOLATILITY MODEL

We are going to introduce the Local Volatility Model mainly following [3, Ch. 2].

6.1 Model description

The **local volatility model** is the simplest and most widely used volatility model in derivatives pricing. It extends the Black-Scholes framework by allowing the instantaneous volatility to be a deterministic function of both time and the spot price of the underlying asset,

$$\sigma = \sigma(t, S).$$

So the dynamics of the underlying asset under the local volatility model are given by the following stochastic differential equation,

$$\frac{dS_t}{S_t} = (r - q)dt + \sigma(t, S_t)dW_t,$$

and the pricing PDE is identical to the Black-Scholes PDE, with the deterministic volatility function replacing the constant volatility

$$\frac{\partial f}{\partial t} = rf(t, S) - (r - q)S \frac{\partial f}{\partial S} - \frac{1}{2}\sigma^2(t, S)S^2 \frac{\partial^2 f}{\partial S^2},$$

6.1.1 Market model

A market model is a valuation model that includes **vanilla options**¹ as part of the underlying, this means we are assuming that when hedging a derivative, we can use the vanilla options as

¹An option is said to be vanilla if their terms are standard and widely quoted in the market, for example European options or American options.

a hedging instrument, in other words, the derivative can be replicated using a combination of the underlying as well as vanilla options. Thus vanilla options under a market model are exactly priced.

The local volatility model is a market model, since it is calibrated to match the vanilla option prices observed in the market.

6.2 Dupire Equation

The Dupire equation, introduced in [10], expresses the local volatility function $\sigma(t, S)$ directly in terms of derivatives of call option prices with respect to strike and maturity. As we have said previously, we present the derivation following [3].

6.2.1 Derivation

Consider the following general diffusive dynamics for S_t :

$$(6.1) \quad dS_t = (r - q)S_t dt + \sigma_t S_t dW_t,$$

where σ_t is an arbitrary adapted process. The price of a European call option with strike K and maturity T follows from the pricing formula (5.6), with payoff $f(S_T) = (S_T - K)^+$ and a constant rate r ,

$$C(K, T) = e^{-rT} \mathbb{E}[(S_T - K)^+],$$

the expectation being taken under the risk-neutral measure $\mathbb{Q}$.

Before applying Itô's formula we recall the two generalized functions that appear as distributional derivatives of $(x)^+ = \max(x, 0)$. Both are understood in the sense of distributions, that is, as continuous linear functionals on the space of test functions $\mathcal{D}(\mathbb{R}) = C_c^\infty(\mathbb{R})$, the smooth functions on $\mathbb{R}$ with compact support; see [13, Ch. 1 and 3] for a rigorous treatment.

Definition 6.1 (Heaviside function). The **Heaviside function** $\theta : \mathbb{R} \rightarrow \{0, 1\}$ is defined by

$$\theta(x) = \begin{cases} 1 & x > 0, \\ 0 & x \leq 0. \end{cases}$$

The value assigned at $x = 0$ is immaterial in what follows, since θ only enters through integrals and $\{0\}$ has zero Lebesgue measure.

Definition 6.2 (Dirac delta). The **Dirac delta** $\delta \in \mathcal{D}'(\mathbb{R})$ is the distributional derivative of θ , characterized by the property that

$$\int_{-\infty}^{+\infty} \varphi(x) \delta(x) dx = \varphi(0), \quad \forall \varphi \in \mathcal{D}(\mathbb{R}),$$

where the integral is the customary notation for the action of δ on φ .

These two distributions arise as the successive generalized derivatives of $(x)^+$:

$$(6.2) \quad \frac{d}{dx}(x)^+ = \theta(x), \quad \frac{d^2}{dx^2}(x)^+ = \delta(x).$$

The first identity follows from $(x)^+ = x\theta(x)$ and $x\delta(x) = 0$; the second is the distributional derivative of θ .

To relate option prices to the dynamics of S_t , we apply Itô's formula to $(S_T - K)^+$ over the interval $[T, T + dT]$, using (6.2) with $x = S_T - K$:

$$\begin{aligned} d(S_T - K)^+ &= \theta(S_T - K)((r - q)S_T dT + \sigma_T S_T dW_T) \\ &\quad + \frac{1}{2} \delta(S_T - K) \sigma_T^2 S_T^2 dT. \end{aligned}$$

We work with undiscounted prices $\mathcal{C}(K, T) = e^{rT} C(K, T)$. Differentiating the identity $\mathcal{C}(K, T) = \mathbb{E}[(S_T - K)^+]$ with respect to K gives:

$$(6.3) \quad \mathbb{E}[\theta(S_T - K)] = -\frac{\partial \mathcal{C}}{\partial K}, \quad \mathbb{E}[\delta(S_T - K)] = \frac{\partial^2 \mathcal{C}}{\partial K^2}.$$

The second identity is the Breeden-Litzenberger result [6]: the second strike derivative of undiscounted call prices equals the risk-neutral density of S_T . To see why, let $p^*(\cdot, T)$ denote the risk-neutral density of S_T , so that

$$\mathcal{C}(K, T) = \mathbb{E}[(S_T - K)^+] = \int_0^\infty (s - K)^+ p^*(s, T) ds = \int_K^\infty (s - K) p^*(s, T) ds.$$

Differentiating with respect to K (using Leibniz's rule, noting the lower limit contributes zero because $(K - K)^+ = 0$):

$$\frac{\partial \mathcal{C}}{\partial K} = - \int_K^\infty p^*(s, T) ds = -(1 - F(K, T)),$$

where $F(K, T) = \mathbb{Q}(S_T \leq K)$ is the risk-neutral CDF. A second differentiation gives:

$$(6.4) \quad \frac{\partial^2 \mathcal{C}}{\partial K^2} = p^*(K, T).$$

Since $\mathbb{E}[\delta(S_T - K)] = \int_0^\infty \delta(s - K) p^*(s, T) ds = p^*(K, T)$, this is exactly the second identity in (6.3). In particular, the positivity of p^* guarantees $\partial^2 \mathcal{C} / \partial K^2 \geq 0$, which we will use when discussing no-arbitrage conditions.

From the identity $\mathcal{C} = \mathbb{E}[S_T \theta(S_T - K)] - K \mathbb{E}[\theta(S_T - K)]$ we also obtain:

$$\mathbb{E}[S_T \theta(S_T - K)] = \mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K}.$$

Taking the expectation of the Itô expansion (noting $\mathbb{E}[dW_T] = 0$) and substituting these identities:

$$\begin{aligned} \frac{\partial \mathcal{C}}{\partial T} dT &= (r - q) \mathbb{E}[S_T \theta(S_T - K)] dT + \frac{1}{2} \mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] dT \\ &= (r - q) \left(\mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K} \right) dT + \frac{1}{2} \mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] dT. \end{aligned}$$

For the remaining term, we use the sifting property of the Dirac delta: since $\delta(S_T - K)$ is supported at $S_T = K$, the factor S_T^2 can be replaced by K^2 inside any expectation,

$$\mathbb{E}[\sigma_T^2 S_T^2 \delta(S_T - K)] = K^2 \mathbb{E}[\sigma_T^2 \delta(S_T - K)],$$

giving:

$$\frac{\partial \mathcal{C}}{\partial T} dT = (r - q) \left(\mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K} \right) dT + \frac{K^2}{2} \mathbb{E}[\sigma_T^2 \delta(S_T - K)] dT.$$

Solving for $\mathbb{E}[\sigma_T^2 \delta(S_T - K)]$, dividing by $\mathbb{E}[\delta(S_T - K)] = \partial^2 \mathcal{C} / \partial K^2$, and reverting to discounted prices $C = e^{-rT} \mathcal{C}$, we obtain:

$$(6.5) \quad \mathbb{E}[\sigma_T^2 | S_T = K] = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}}.$$

This identity holds for any model of the lognormal-type form (6.1), with σ_t an arbitrary adapted process [3]. The multiplicative structure $\sigma_t S_t$ is essential: it is what produces the factor S_T^2 in the Itô expansion, which the sifting property then converts to K^2 . A consequence is that the smile alone does not determine σ_t : two processes σ_t and σ'_t generate the same vanilla smile if and only if $\mathbb{E}[\sigma_T^2 | S_T = K] = \mathbb{E}[\sigma_T'^2 | S_T = K]$ for all K and T .

In the local volatility model, where $\sigma_t \equiv \sigma(t, S_t)$, the conditional expectation reduces to $\sigma(t, S)^2$, and evaluating at $K = S$, $T = t$ gives the **Dupire formula**:

$$(6.6) \quad \sigma(t, S)^2 = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}} \Big|_{\substack{K=S, \\ T=t}}.$$

Given the full surface of observed call prices $C(K, T)$, equation (6.6) uniquely determines the local volatility function.

6.2.2 Forward equation

In the local volatility model the left hand side of (6.6) is $\sigma^2(T, K)$, so for a known local volatility function that same identity can be rearranged into the **Dupire forward equation**:

$$(6.7) \quad \frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} - \frac{\sigma^2(T, K)}{2} K^2 \frac{\partial^2 C}{\partial K^2} = -qC, \quad (K, T) \in (0, \infty) \times (0, T_{\max}],$$

where $T_{\max}$ is the longest maturity of interest. Unlike the Black-Scholes backward PDE, which is solved backwards in t from a terminal payoff for a single strike and maturity, the forward equation propagates forward in T with initial condition $C(K, T = 0) = (S_0 - K)^+$, producing prices for a fixed initial spot S_0 but all strikes K and maturities T simultaneously.

6.2.3 No-arbitrage conditions

Absence of arbitrage in the option surface makes both the numerator and the denominator of (6.5) non-negative. The Dupire formula (6.6) then yields a non-negative local variance at every (K, T) where the denominator is strictly positive, that is, wherever the risk-neutral density of S_T does not vanish.

Strike arbitrage. From (6.3), the denominator satisfies:

$$\frac{\partial^2 C(K, T)}{\partial K^2} = e^{-rT} \mathbb{E}[\delta(S_T - K)] \geq 0,$$

since it is proportional to the risk-neutral density of S_T .

We can also get to the same condition by contradiction. Suppose $\partial^2 C / \partial K^2 < 0$ at some (K, T) and consider a **butterfly spread**², buying ε^{-2} calls at strike $K - \varepsilon$, buying ε^{-2} calls at strike $K + \varepsilon$, and selling $2\varepsilon^{-2}$ calls at strike K . Its value at inception, the premium paid to enter the position, is the second difference

$$V_0(\varepsilon) = \frac{C(K - \varepsilon, T) - 2C(K, T) + C(K + \varepsilon, T)}{\varepsilon^2} \xrightarrow{\varepsilon \rightarrow 0} \frac{\partial^2 C}{\partial K^2} < 0,$$

so for ε small enough we are paid to enter the position. Its payoff at maturity,

$$\frac{(S_T - K + \varepsilon)^+ - 2(S_T - K)^+ + (S_T - K - \varepsilon)^+}{\varepsilon^2} \geq 0,$$

is non-negative for any realisation of S_T by convexity of $x \mapsto (x)^+$. This is an arbitrage in the sense of Definition 5.5, so $\partial^2 C / \partial K^2 \geq 0$.

Maturity arbitrage. The numerator of (6.5) can be rewritten as:

$$e^{-qT} \frac{d}{dT} \left[e^{qT} C(K e^{(r-q)T}, T) \right],$$

which is non-negative if and only if e^{qT} times the price of a call struck at a fixed proportion of the forward $F_T = S_0 e^{(r-q)T}$ is non-decreasing in maturity. A violation constitutes a **maturity arbitrage**: if the longer-dated call is worth less than the shorter-dated one, buying the former and selling the latter is entered at a negative premium, and holding the forward position that matches the two strikes leaves a non-negative profit and loss (P&L) at both maturities, which is again an arbitrage in the sense of Definition 5.5.

In an arbitrage-free market both conditions hold and the Dupire local volatility function is well-defined.

²A butterfly spread is the combination of a long call at a low strike, a long call at a high strike, K , and two short calls at the strike in between, $K \pm a$, all with the same maturity, thus the payoff is 0 if $S_T \leq K - a$ or $S_T \geq K + a$, and positive otherwise with a maximum payoff at $S_T = K$. It is the standard way of taking a position on the curvature of the price surface in the strike direction.

One remark is worth making before leaving this point. Both conditions are conditions on the *prices*, so any arbitrage-free surface admits a local volatility. In particular the steep skews observed at short maturities on equity indices are no evidence that jumps are needed to produce them [3, Ch. 2].

6.3 From local volatilities to implied volatilities

The Dupire formula (6.6) gets the local volatility from the option prices. In the opposite direction, how a given local volatility function shows up in the implied volatility surface, a.k.a. the smile, has no closed form, but admits approximations that are accurate enough to explain the shape of the smile the model produces and, more importantly, how that smile moves. We collect here the results we use later, following [3, Ch. 2] and [9].

6.3.1 Implied volatility as an average of local volatility

Let $C_{\hat{\sigma}}$ be the Black-Scholes price of a call option with strike K and maturity T computed with a constant volatility equal to its implied volatility $\hat{\sigma}(K, T)$ as defined in (4.9). Delta hedging this option with the Black-Scholes model, i.e. using the implied volatility, while the spot actually diffuses with the local volatility $\sigma(t, S)$ generates, between two rehedges, the gamma/theta non-zero P&L given by

$$\frac{1}{2} S_t^2 \frac{\partial^2 C_{\hat{\sigma}}}{\partial S^2} (\sigma(t, S_t)^2 - \hat{\sigma}^2) dt.$$

By construction the two models price the option identically, so the discounted P&L over time has zero expected value under the local volatility dynamics, which after rearranging gives [3, Ch. 2]

$$(6.8) \quad \hat{\sigma}(K, T)^2 = \frac{\mathbb{E} \left[\int_0^T e^{-rt} S_t^2 \frac{\partial^2 C_{\hat{\sigma}}}{\partial S^2} \sigma(t, S_t)^2 dt \right]}{\mathbb{E} \left[\int_0^T e^{-rt} S_t^2 \frac{\partial^2 C_{\hat{\sigma}}}{\partial S^2} dt \right]},$$

the expectation being taken over paths of the local volatility model.

Thus implied variance, $\hat{\sigma}^2$, is an average of the local variance, $\sigma(t, S)^2$, along the paths, weighted by the gamma of the option. The relation is implicit, since $\hat{\sigma}$ appears in the weights on the right, so it is not a practical pricing formula, but it gives the intuition that an implied volatility is not a local volatility at a point, but a smoothed version of it over the region of (t, S) where the option carries gamma. The smile is therefore always flatter than the local volatility surface that generates it.

6.3.2 The rule of two

A central result of the local volatility model concerns the relationship between the skew of the local volatility function and the skew of the implied volatility smile.

Define the at the money forward, ATMF, skew $\mathcal{S}_T$ as the slope of the implied volatility smile with respect to log-strike, evaluated at the forward $F_T = S_0 e^{(r-q)T}$:

$$(6.9) \quad \mathcal{S}_T := \left. \frac{\partial \hat{\sigma}(K, T)}{\partial \ln K} \right|_{K=F_T}.$$

Consider a local volatility function that is, near the forward, linear in log-moneyness

$$(6.10) \quad \sigma(t, S) = \sigma_0 + \alpha(t)x + \dots, \quad x = \ln\left(\frac{S}{F_t}\right),$$

where $\sigma_0 = \sigma(0, S_0)$ is the level of the local volatility and $\alpha(t)$ is its skew at time t .

To first order in α , the implied ATMF skew is given by [3, Ch. 2]

$$(6.11) \quad \mathcal{S}_T = \frac{1}{T} \int_0^T \frac{t}{T} \alpha(t) dt.$$

If $\alpha(t) = \alpha$ is constant in time, the integral gives

$$(6.12) \quad \mathcal{S}_T = \frac{1}{2} \alpha.$$

Thus, for a constant local volatility skew, the implied skew is half the local skew. The averaging inherent in the mapping from local to implied volatilities smooths out the local volatility surface, making the implied smile flatter.

6.3.3 Smile dynamics and the skew stickiness ratio

In the local volatility model, the smile's dynamics are fully determined by the smile itself. There is no freedom to choose how implied volatilities move when the spot moves.

The key quantity is the ATMF volatility $\hat{\sigma}_{F_T} := \hat{\sigma}(F_T, T)$, the implied volatility for a strike equal to the forward, and the question is how it changes when the spot moves by $d \ln S_0$.

To first order in the local skew $\alpha(t)$ of (6.10),

$$(6.13) \quad \frac{d\hat{\sigma}_{F_T}(T)}{d \ln S_0} = \frac{1}{T} \int_0^T \alpha(t) dt.$$

Using (6.11) to write α in terms of the ATMF skew, this becomes [3, Ch. 2]

$$(6.14) \quad \frac{d\hat{\sigma}_{F_T}(T)}{d \ln S_0} = \mathcal{S}_T + \frac{1}{T} \int_0^T \mathcal{S}_t dt.$$

The rate at which the ATMF volatility moves with the spot is thus dictated entirely by the term structure of the ATMF skew.

Practitioners define the **skew stickiness ratio** (SSR) as

$$(6.15) \quad \mathcal{R}_T := \frac{1}{\mathcal{S}_T} \frac{d\hat{\sigma}_{F_T}(T)}{d \ln S_0}.$$

The SSR measures how much the ATMF volatility moves with the spot, in units of the ATMF skew. Two familiar regimes serve as benchmarks:

- $\mathcal{R}_T = 0$: the smile moves with the spot so that volatilities at fixed log-moneyness stay constant (sticky-delta).
- $\mathcal{R}_T = 1$: the smile stays fixed in strike space, so the ATMF volatility slides along the existing smile (sticky-strike).

In the local volatility model, substituting (6.14) into (6.15),

$$(6.16) \quad \mathcal{R}_T = 1 + \frac{1}{T} \int_0^T \frac{\mathcal{S}_t}{\mathcal{S}_T} dt.$$

If the skew is constant across maturities ($\mathcal{S}_t = \mathcal{S}_T$ for all t), then $\mathcal{R}_T = 2$ for every maturity.

In summary:

In the local volatility model, the spot/volatility covariance is not a free parameter. It is fixed by the market smile used for calibration. The SSR quantifies this covariance in units of the skew.

CALIBRATION OF LOCAL VOLATILITY

The local volatility model must be calibrated to market option prices to be usable. Given quoted call prices $C^*(K_i, T_j)$ for a finite set of strikes and maturities, we seek a function $\sigma(S, t)$, or equivalently $a(y, \tau) = \frac{1}{2}\sigma^2$, such that the Dupire forward equation (6.7) reproduces the observed prices. Written out, the direct problem we invert is

$$\frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} - \sigma^2(K, T) \frac{K^2}{2} \frac{\partial^2 C}{\partial K^2} + qC = 0,$$

for $(K, T) \in (0, \infty) \times (0, T_{\max}]$, with initial condition $C(K, 0) = (S_0 - K)^+$ and the boundary behaviour $C \rightarrow S_0 e^{-qT}$ as $K \rightarrow 0$ and $C \rightarrow 0$ as $K \rightarrow \infty$. The unknown is the coefficient σ , the data are values of the solution C .

We follow [11] and treat the calibration as a Tikhonov regularization problem.

7.1 Ill-posedness of the inverse pricing problem

The calibration of the local volatility model is an ill-posed inverse problem, because on one hand market data is noisy and sparse in maturity. On the other hand the map from a to option prices has an smoothing effect that makes its inverse discontinuous. This is important, it means different volatility functions can produce almost the same option prices, especially away from the money.

We can see the latter in the Dupire equation (6.5), the gamma is in the denominator, so in regions where gamma is small, like far OTM/ITM, the local volatility is not well determined by the prices.

Also the Local Volatility model assumes that option prices are once continuously differentiable in T and twice in K , we are looking for a function $\sigma(S, t)$ that reproduces the market prices, but

PDE theory suggests that the coefficient σ should live in a more general space, like H^1 or L^2 . This arises the problem, where the solution is smoother than the coefficient.

7.2 Change of variable - log-moneyness

Similar to what we did in Section 4.3 for the Black-Scholes PDE, we change variables to log-moneyness

$$y = \log(K/S_0), \quad \text{equivalently} \quad K = S_0 e^y,$$

removes this dependence and puts the PDE in a translation-invariant form, just as the substitution $x = \log(S/K)$ reduced the Black-Scholes PDE to the heat equation in Section 4.3. Define the time to maturity $\tau = T$, the diffusion coefficient

$$a(y, \tau) = \frac{1}{2} \sigma^2(K, T),$$

and the normalized call price

$$u(y, \tau) = C(S_0; K, T)/S_0,$$

so that $C(S_0; K, T) = S_0 u(y, \tau)$.¹

We now carry the Dupire forward equation (6.7) into the (y, τ) coordinates. Since $y = \log(K/S_0)$ we have $\partial y / \partial K = 1/K$, and because $\tau = T$ the maturity derivative is unchanged. Applying the chain rule to $C = S_0 u(y, \tau)$ gives the three derivatives that appear in (6.7):

$$\begin{aligned} \frac{\partial C}{\partial T} &= S_0 u_\tau, \\ \frac{\partial C}{\partial K} &= S_0 u_y \frac{\partial y}{\partial K} = \frac{S_0}{K} u_y, \\ \frac{\partial^2 C}{\partial K^2} &= \frac{\partial}{\partial K} \left(\frac{S_0}{K} u_y \right) = \frac{S_0}{K} u_{yy} \frac{\partial y}{\partial K} - \frac{S_0}{K^2} u_y = \frac{S_0}{K^2} (u_{yy} - u_y). \end{aligned}$$

The convenient feature of log-moneyness is that the first-order operator $K \partial_K$ and the second-order operator $K^2 \partial_{KK}$ both become constant-coefficient operators in y :

$$K \frac{\partial C}{\partial K} = S_0 u_y, \quad K^2 \frac{\partial^2 C}{\partial K^2} = S_0 (u_{yy} - u_y),$$

which is exactly what makes the equation translation-invariant in y . Substituting these into (6.7) and using $a = \frac{1}{2} \sigma^2$,

$$S_0 u_\tau + (r - q) S_0 u_y - a S_0 (u_{yy} - u_y) = -q S_0 u.$$

Dividing through by S_0 and solving for u_τ yields the forward equation in log-moneyness form,

$$(7.1) \quad u_\tau = a(y, \tau) (u_{yy} - u_y) - (r - q) u_y - q u,$$

¹We keep the explicit zero-order term. [11] instead set $u = e^{\int_t^T q ds} C$, which absorbs the dividend factor and yields the equivalent equation $-u_\tau + a(u_{yy} - u_y) + (q - r)u_y = 0$ without a $-qu$ term; the two normalizations coincide when $q = 0$, as in Example 1 below. We implemented and tested both on the market surface, where $q \neq 0$, and neither outperformed the other; see Appendix B.12.

on the truncated domain $Q_{M,T} = (-M, M) \times (0, T]$. The initial condition follows from the same normalization applied to the call payoff $C(K, 0) = (S_0 - K)^+$:

$$u(y, 0) = \frac{(S_0 - K)^+}{S_0} = \left(1 - \frac{K}{S_0}\right)^+ = (1 - e^y)^+,$$

that is,

$$(7.2) \quad u(y, 0) = \max(1 - e^y, 0)$$

and Dirichlet boundary conditions

$$(7.3) \quad u(-M, \tau) = \Psi_1(\tau), \quad u(M, \tau) = \Psi_2(\tau).$$

The boundary values are determined from the Black-Scholes formula evaluated at the boundary strikes $K = S_0 e^{\pm M}$: as $M \rightarrow \infty$, the left boundary ($K \rightarrow 0$) gives the intrinsic value $\Psi_1(\tau) \rightarrow e^{-q\tau}$, and the right boundary ($K \rightarrow \infty$) gives $\Psi_2(\tau) \rightarrow 0$.

7.3 Parameter-to-solution map

Before stating the admissible set we fix the notation for the spaces involved, following [7, Ch. 8 and 9] for the general theory. On a domain D we write $L^p(D)$ for the Lebesgue space of p -integrable functions, with $L^2(D)$ a Hilbert space for the inner product and norm

$$(f, g)_{L^2(D)} = \int_D f g, \quad \|f\|_{L^2(D)}^2 = (f, f)_{L^2(D)},$$

and $L^\infty(D)$ for the essentially bounded functions. The Sobolev space $H^1(D) = W^{1,2}(D)$ collects the $f \in L^2(D)$ whose first order weak derivatives are again in $L^2(D)$, normed by

$$\|f\|_{H^1(D)}^2 = \|f\|_{L^2(D)}^2 + \|\nabla f\|_{L^2(D)}^2,$$

and $H_0^1(D)$ is the closure of the smooth compactly supported functions in $H^1(D)$, the subspace vanishing on ∂D . We also use the parabolic space $W_p^{2,1}(Q)$ of functions on the space-time domain Q with two weak spatial and one weak time derivative in $L^p(Q)$. All derivatives below are weak derivatives, written ∂_y in the penalty terms and abbreviated by subscripts, $u_y = \partial_y u$, inside the equations.

For each admissible diffusion coefficient a , let $u(a)$ denote the solution to (7.1) - (7.2). Fix an *a priori* reference $a^* \in L^\infty(\mathbb{R})$ with $0 < \underline{a} \leq a^* \leq \bar{a}$, where $\underline{a}, \bar{a}$ are self-imposed bounds on the volatility, and define the set of admissible parameters

$$(7.4) \quad \mathcal{K}(a^*) := \{a \in a^* + H^1(Q) : \underline{a} \leq a \leq \bar{a}\}.$$

Here $Q = \Omega \times (0, T)$ is the space-time domain and the span space $a^* + H^1(Q)$ is taken such as $\nabla a^* \in [L^2(Q)]^2$ (Definition 2.1 of [11]).

Two features of H^1 matter. As we will present afterwards in (7.13), the coefficient enters the equation through the flux term, so it must have a weak derivative in L^2 . And also the H^1 penalty makes the sublevel sets of the Tikhonov functional bounded in H^1 , hence weakly compact, which is what yields a minimizer. In the synthetic examples below the volatility is time independent, $a = a(y)$, and the penalty reduces to the spatial norm

$$\|a\|_{H^1(\Omega)}^2 = \int_{-M}^M (a^2 + (\partial_y a)^2) dy,$$

on the bounded interval H^1 then also embeds into $C[-M, M]$, so the bounds $\underline{a} \leq a \leq \bar{a}$ hold pointwise.

The *parameter-to-solution map* $F = F_{a_0}$ is defined as

$$(7.5) \quad F(a) = u(a) - u(a_0),$$

where $a_0 \in \mathbb{R}$ with $\underline{a} \leq a_0 \leq \bar{a}$ is a reference constant and $u(a_0)$ can be computed from the Black-Scholes formula. A key regularity result established in [11] is the following.

Theorem 7.1 ([11], Theorem 2.1). *There exists $p^* > 2$ such that $F: \mathcal{K}(a^*) \rightarrow W_p^{2,1}(Q)$ is continuous and compact for $2 \leq p < p^*$, and weakly sequentially continuous.*

The compactness and weak closeness of F is the mathematical source of the ill-posedness. As already mentioned it reflects the smoothing action of the pricing equation, the solution of a parabolic problem is more regular than its coefficient, and the solution space $W_p^{2,1}(Q)$ embeds compactly in the observation space. A compact operator between infinite dimensional spaces cannot have a continuous inverse, so arbitrarily small changes in the observed prices can correspond to order one changes in a . In practice very different volatility profiles reproduce almost the same option prices, especially away from the money, and any direct inversion amplifies the noise.

Controlling this instability is the purpose of the Tikhonov penalty.

7.4 Tikhonov functional

Lets assume that there exists $a^\dagger \in \mathcal{K}(a^*)$ such that $u^\dagger = u(a^\dagger)$ reproduces the 'true' prices $u^\dagger$ exactly. And let u^δ denote the noisy observed option prices with a bound known a priori,

$$\|u^\delta - u^\dagger\|_{L^2(Q)} \leq \delta.$$

Following [11] we stabilize the inverse problem by minimizing the *Tikhonov functional*

$$(7.6) \quad J^\beta(a) = \|u(a) - u^\delta\|_{L^2(Q)}^2 + \beta \|a - a^*\|_{H^1(\Omega)}^2,$$

where $\beta > 0$ is the regularization parameter, and a^* is the *a priori* guess that encodes any structural knowledge about the volatility (e.g., smoothness or smile shape). The first term

penalizes any mismatch with the data, the second term penalizes the deviation of a from a^* in H^1 , which promotes smoother reconstructions.

with $\Omega = (-M, M)$ the spatial domain, so that the penalty reads

$$\|a - a^*\|_{H^1(\Omega)}^2 = \int_{-M}^M [(a - a^*)^2 + (\partial_y(a - a^*))^2] dy.$$

When a is genuinely time dependent the same norm is taken on $Q = \Omega \times (0, T)$, with ∇ in place of ∂_y .

Proposition 7.2 ([11], Propositions 3.1 and 3.2). *For any $\beta > 0$ a minimizer a_β^δ of J^β exists. If $\delta \rightarrow 0$ and $\beta(\delta) \rightarrow 0$ with $\delta^2/\beta(\delta) \rightarrow 0$, then every sequence of minimizers $\{a_{\beta(\delta_k)}^{\delta_k}\}$ has a subsequence converging in $H^1(\Omega)$, and every limit point is an a^* -minimum-norm solution, that is, a solution of the noise free problem at least H^1 distance from the prior a^* .*

An important remark is done in [11]. Usually in the analysis of the inverse problems, in particular with the Tikhonov regularization approach, is assumed that the observations of the solutions of the PDE are continuous, in our case we would like to have as data the prices of the options in any strike or maturity, as a continuous function. But in practice we only have a finite number of prices, so the data is discrete. This makes the convergence theory more complicated, in particular the convergence to some a^* -minimum-norm solution is guaranteed, but its uniqueness is not, see [12, Ch. 10 and 11] for the general theory behind this point.

7.5 Gradient computation via the adjoint method

Before tackling the convergence of the minimization method, we need to compute the gradient of the Tikhonov functional J^β , in particular we need its Gâteaux derivative. This is the *adjoint method*, Section 5.1 of [11], and it is worth saying up front what it buys us. The derivative of J^β at a is a directional derivative, one number per direction h , and a discretization with n degrees of freedom has n basis directions, so reading the gradient off the definition costs n linearized solves. The adjoint method instead identifies the gradient with a single function, computed from three parabolic problems, the forward equation (7.1) for $u(a)$, its linearization (7.7) for $u'(a)[h]$, and the backward adjoint equation (7.8) for V , of which only the first and the last are needed per iteration. The three are implemented in `pde/forward.py`, `pde/linearized.py` and `pde/adjoint.py`, see Appendix C.

The *directional derivative* of the data-fidelity term at a in direction h is

$$\frac{d}{d\varepsilon} \|u(a + \varepsilon h) - u^\delta\|_{L^2(Q)}^2 \Big|_{\varepsilon=0} = 2(u(a) - u^\delta, u'(a)[h])_{L^2(Q)},$$

which is the chain rule applied to the quadratic $\|\cdot\|_{L^2}^2$, and where $u'(a)[h] = w(\cdot, T_0)$ is the solution of the *linearized* forward equation, obtained by differentiating (7.1) in the direction h ,

$$(7.7) \quad -w_\tau + a(w_{yy} - w_y) - (r - q)w_y - qw = -h(u_{yy} - u_y), \quad w(y, 0) = 0, \quad w|_{\partial\Omega} = 0.$$

Rather than solving (7.7) for every direction h , the gradient is evaluated efficiently via a single *adjoint solve*. Let $V(y, \tau)$ solve the backward parabolic problem

$$(7.8) \quad V_\tau + (aV)_{yy} + (aV)_y + (r - q)V_y - qV = 0, \quad V(y, T_0) = u(a)(y, T_0) - u^\delta(y),$$

with homogeneous Dirichlet conditions at $y = \pm M$ and integrated backward in τ . Then the Gâteaux derivative satisfies [11]

$$(u'(a)[h], u(a) - u^\delta)_{L^2(Q)} = \int_0^{T_0} \int_{-M}^M h(y)(u_{yy} - u_y)V dy d\tau.$$

The regularization term, is an explicit functional of a , so its Gâteaux derivative needs no adjoint, we find it by differentiating the quadratic form directly gives

$$\frac{d}{d\varepsilon} \|a + \varepsilon h - a^*\|_{H^1(\Omega)}^2 \Big|_{\varepsilon=0} = 2(a - a^*, h)_{H^1(\Omega)} = 2 \int_{-M}^M [(a - a^*)h + \partial_y(a - a^*)\partial_y h] dy.$$

Adding this to the data term above, the total gradient direction (in L^2 sense) is

$$(7.9) \quad \langle \nabla J^\beta(a), h \rangle = 2 \int_0^{T_0} \int_{-M}^M h(u_{yy} - u_y)V dy d\tau + 2\beta \int_{-M}^M [(a - a^*)h + \partial_y(a - a^*)\partial_y h] dy.$$

7.6 Convergence rates and the source condition

The adjoint calculation of the previous section computes the action of $F'(a)^*$ on the residual. Writing the data discrepancy as $\|F(a) - u^\delta\|^2$ and using

$$\langle \nabla J_d(a), h \rangle = 2(F'(a)h, F(a) - u^\delta) = 2(h, F'(a)^*(F(a) - u^\delta)),$$

the solution V of (7.8) represents $F'(a)^*(F(a) - u^\delta)$, see [11, Remark 4.2]. The same adjoint controls the speed of convergence.

The convergence results exposed in proposition 7.2 can be quite slow, they give no rate of convergence. Without a further assumptions we can no derive such a rate of convergence. The general theory [12] provides a rate under a source condition on the true coefficient,

$$(7.10) \quad a^* - a^\dagger = F'(a^\dagger)^* w,$$

holds for small $\|w\|$. And F is Frechet differentiable and F' is Lipschitz continuous in a neighborhood of $a^\dagger$. Though in [11] is mentioned that F just needs to be Gateaux differentiable and Lipschitz continuous in some directions, so we will be able to use the Gateaux derivative we just computed.

By [11, Proposition 4.1], for $\|w\|$ small enough, the choice $\beta \sim \delta$ gives the convergence rates

$$(7.11) \quad \|a_\beta^\delta - a^\dagger\| = O(\sqrt{\delta}), \quad \|u(a_\beta^\delta) - u^\delta\| = O(\delta).$$

The abstract condition (7.10) is hard to check directly, so Egger and Engl derive interpretable sufficient conditions by restricting to a single maturity and a time independent volatility, $a(\tau, y) =$

$a(y)$. In that setting they prove two rate results of the form (7.11). First, [11, Theorem 4.1] treats observations of prices, with the penalty strengthened to the $H^2(\Omega)$ norm. Second, [11, Theorem 4.2] keeps the $H^1(\Omega)$ penalty but assumes that the derivative of the option price is also observable, which is realistic because $C_K = e^{-y}u_y$ is the price of a digital call and is quoted on the market.

In both, the source function must be smooth and decay like a Gaussian at both ends. Writing $g = \partial_y^4 w - \partial_y^2 w + w$ for the first result and $g = -\partial_y^2 w + w$ for the second, the requirement is

$$(7.12) \quad |g|, |\partial_y g|, |\partial_y^2 g| = O(e^{-B'y^2}) \quad \text{as } |y| \rightarrow \infty,$$

for some $B' > 0$.

This decay condition is the precise form of the wing dynamics, and Egger and Engl argue the same in their concluding remarks, option prices in a sense reflect the market's expectation of future volatility, which is reasonably smooth, but for deep in-the-money/ out-of-the-money options, volatility is too noisy given the data, so that structural behavior must come from the prior a^* [11]. The condition requires $a^\dagger - a^*$ to be small and smooth in the wings, which is why the priors in the experiments below are chosen to flatten away from the money.

Two points carry over to the rest of the chapter, and neither is proved by the theorems above. The functional we minimize keeps the H^1 penalty with price data only, which is the setting of the stability and convergence results but not literally of either rate theorem, since those need the stronger $H^2(\Omega)$ penalty or the added derivative data. And the rate analysis is proved only for a time independent volatility, so the genuinely time dependent surface $a(y, \tau)$ we calibrate in Section 7.11 lies outside its scope.

7.7 Choice of β and stopping criterion

In the numerical experiments of [11] the regularization parameter is chosen by a decreasing sequence

$$\beta_0 = 10^2 \delta > \beta_1 > \dots > \beta_n = 10^{-3} \delta,$$

minimizing J^{β_i} sequentially, starting each minimization step from the previous minimizer solution.

The sequence is stopped when the Morozov discrepancy principle

$$\|u(a_\beta^\delta) - u^\delta\| \leq \tau \delta$$

is satisfied for some $\tau > 1$. In [11] they mention that after testing across the noise levels $\delta \in [10^{-5}, 10^{-1}]$, the choice $\tau = 1.5$ could be met at typical values $\beta \sim 10^{-2} \delta$. For exact data (i.e. no noise, $\delta = 0$) a single fixed β is used.

7.8 Variational forms and discretization

The forward PDE (7.1) is discretized in space with P_1 Lagrange finite elements on a uniform mesh of $[-M, M]$ and in time with the implicit Euler scheme. Both are assembled and solved with FEniCSx [1], the mesh, the element degree and the domain truncation M are collected in Appendix B.1, and the solvers are `pde/forward.py` and `pde/adjoint.py` of Appendix C. Multiplying by a test function $v \in H_0^1(-M, M)$ and integrating by parts gives the variational form

$$(7.13) \quad \frac{d}{d\tau} \int_{-M}^M u v dy = - \int_{-M}^M a u_y v_y dy - \int_{-M}^M \partial_y a u_y v dy - \int_{-M}^M (a + r - q) u_y v dy - \int_{-M}^M q u v dy.$$

The adjoint equation (7.8) is run backward in τ , again multiplying by a test function $v \in H_0^1(-M, M)$ and integrating by parts gives its weak form (with $m = V$)

$$(7.14) \quad \frac{d}{d\tau} \int_{-M}^M m v dy = \int_{-M}^M (a m)_y v_y dy + \int_{-M}^M a m v_y dy + \int_{-M}^M (r - q) m v_y dy + \int_{-M}^M q m v dy.$$

7.8.1 Time discretization

We discretize of time by N arbitrary small steps $\tau_0, \tau_1, \dots, \tau_N$ with the implicit Euler scheme, which is defined by

$$(7.15) \quad \frac{1}{\Delta\tau} \int_{-M}^M (u^n - u^{n-1}) v dy = \phi(a^n; u^n, v) \quad \text{for all } v \in V_h \cap H_0^1(-M, M),$$

for $n = 1, \dots, N$, where $u^n = u(\cdot, \tau_n)$, $\Delta\tau = \tau_n - \tau_{n-1}$, and ϕ is the right-hand side of (7.13),

$$\phi(a^n; u^n, v) = - \int_{-M}^M a^n u_y^n v_y dy - \int_{-M}^M (\partial_y a^n) u_y^n v dy - \int_{-M}^M (a^n + r - q) u_y^n v dy - \int_{-M}^M q u^n v dy.$$

This gives a linear system for each step, is unconditionally stable for this class of parabolic operators, see [22].

7.8.2 A time-dependent coefficient

The admissible set (7.4) is stated on the space $Q = \Omega \times (0, T)$, so the theory shown in Proposition 7.2 covers time-dependent coefficients. However, as previously stated, in practice option prices are observed at finitely many maturities $\tau_1 < \dots < \tau_N$, and anything finer than the gaps between quoted maturities is not identifiable. We therefore choose to model a as piecewise constant in time, on $K \leq N$ intervals whose edges are taken among the observed maturities,

$$(7.16) \quad a(y, \tau) = \sum_{n=1}^K a_n(y) \chi_{(\tau_{n-1}, \tau_n]}(\tau), \quad a_n \in H^1(-M, M), \quad \underline{a} \leq a_n \leq \bar{a},$$

for $n = 1, \dots, K$ with $\tau_0 = 0$. In the experiments below we take one slice per observed maturity, $K = N$. The unknown is the stack of spatial slices $(a_1, \dots, a_K)$, and $K = 1$ recovers the time-independent setting of [11]. The Tikhonov functional is adapted accordingly

$$(7.17) \quad J(a) = \sum_{j=1}^N \|u(a)(\cdot, \tau_j) - u_j^\delta\|_{L^2(\Omega_j)}^2 + \beta_y \sum_{n=1}^K \|a_n - a_n^*\|_{H^1(\Omega)}^2 + \beta_\tau \sum_{n=2}^K \|a_n - a_{n-1}\|_{L^2(\Omega)}^2,$$

where Ω_j is the observed strike band at maturity τ_j . The H^1 term smooths in strike, the new L^2 term penalizes jumps between consecutive slices.

7.9 Minimization Algorithm

We minimize the discretized Tikhonov functional with L-BFGS-B [8], a quasi-Newton method that handles the bound constraints $0 < \underline{a} \leq a_k \leq \bar{a}$. Each iteration requires one forward solve of (7.15) and one backward sweep of the adjoint equation (7.14), which together assemble the gradient (7.9) at the cost of two PDE solves per iteration. Independent of the number of degrees of freedom.

7.9.1 Verification of the discrete adjoint gradient

The adjoint gradient is verified by two tests, both implemented in `derivative_test.py`, see Appendix C.

The first is an exact duality check between the linearized and the adjoint solvers. Write the data-fidelity term as

$$J_d(a) = \|u(a) - u^\delta\|_{L^2(Q)}^2$$

with residual $r = u(a) - u^\delta$. The gradient returned by the adjoint solve should satisfy

$$\langle \nabla J_d(a), h \rangle_{\mathbb{R}^n} = 2(r, u'(a)[h])_{L^2(Q)}$$

for every direction h , with the right-hand side computed separately from the linearized solve (7.7). This equality is exact, with no finite-difference truncation, so if it holds the discrete adjoint is the exact transpose of the linearized forward operator. For a random h on the Example 1 grid the two sides differ by 2.3×10^{-15} , i.e. at the level of rounding error.

The second is a Taylor remainder test on the full functional. When the gradient is correct, the first-order Taylor remainder of the full functional decays one order faster in t than the zeroth-order one

$$R_0(t) = |J(a + th) - J(a)| = O(t), \quad R_1(t) = |J(a + th) - J(a) - t \langle \nabla J(a), h \rangle| = O(t^2).$$

Table 7.1 lists R_0 and R_1 as the step t is reduced geometrically. The remainder R_1 falls at rate ≈ 2 , as it should when ∇J is exact.

7.10 Numerical experiment - Example 1

We reproduce [11, Example 1], implemented in `examples/example1.py`, see Appendix C, where we try to recover a constant volatility on $y \in [-3, 3]$ ($M = 3$), $\tau \in [0, 1]$, with $r = q = 0$ and spot

t	$R_0(t)$	$R_1(t)$	rate of R_1
10^{-2}	1.32×10^{-6}	5.13×10^{-6}	-
10^{-3}	3.28×10^{-7}	5.27×10^{-8}	1.99
10^{-4}	3.76×10^{-8}	5.29×10^{-10}	2.00
10^{-5}	3.80×10^{-9}	5.29×10^{-12}	2.00

Table 7.1: Taylor remainder test for the discrete adjoint gradient (Example 1 setup 7.10). The zeroth-order remainder R_0 decays at first order in t and the first-order remainder R_1 at second order (rate ≈ 2), which is what an exact gradient produces. The rate column is $\log_{10}(R_1(10t)/R_1(t))$.

$S_0 = 1000$.² The true parameter is

$$a^\dagger = 0.15 \iff \sigma^\dagger = \sqrt{2 \times 0.15} \approx 0.548.$$

For part (A) of the example, the data u^δ are the exact option prices at $\tau = T = 1$, generated from the Black-Scholes formula. The initial guess and a^* are both set to

$$a_1^*(y) = 0.15 - 0.05 \operatorname{erf}(-y^2),$$

which agrees with the true value at the money and the wings flatten out ($\partial_y a^* \rightarrow 0$, so $a^* \rightarrow 0.20$) for large $|y|$, the asymptotic behavior invoked in [11] for the convergence-rate theorems 4.1-4.2. The regularization parameter is fixed at $\beta = 10^{-6}$.

Subsequent parts increase the difficulty of the problem, in part B consists on taking 20 strikes only. For part C the prior is set to the deliberately poor constant $a^* = 0.15$. Part D adds 0.1% noise to the continuous data. And finally part BD is both incomplete and noisy.

Our results are shown in following table 7.2

Figures 7.1, 7.2 and 7.3, illustrate how good of a fit we managed to get. As shown the observed region is much more narrow than the domain, we manage to recover a good approximation of the true value in the observed region but on the wings looks like the prior dominates the solution, this is quite clear if we compare figures 7.1 and 7.2.

While in D and BD we can see that the noise in the data makes the recovery of the true value more difficult, we interpreted that the 0.1% noise is applied to the call prices, thus deep ITM options would have a bigger shock in absolute terms than deep OTM options. We don't know if that was Egger's and Engl's approach, as they mention that the noise would be 1\$, as the underlying is 1000\$, but that would probably make some deep OTM options to have negative prices, which is not possible, thus recovering the true value would be impossible in those cases.

²The paper's text states $S_0 = 100$, but its tabulated option values (e.g. a call worth 439 at $K = 600$) and test (D) ("an underlying worth 1000\$") are only consistent with $S_0 = 1000$, we use the latter. The recovered $a(y)$ is scale-invariant, but the Tikhonov balance is not, the data term scales as S_0^2 while the penalty does not, so reproducing the paper's β requires its S_0 .

Strike	True Value	A	B	C	D	BD
600	0.1500	0.1493	0.1500	0.1519	0.1378	0.1245
700	0.1500	0.1498	0.1503	0.1491	0.1386	0.1753
800	0.1500	0.1504	0.1499	0.1503	0.1364	0.1417
900	0.1500	0.1494	0.1499	0.1514	0.1372	0.1589
1000	0.1500	0.1499	0.1499	0.1515	0.1490	0.1455
1100	0.1500	0.1498	0.1500	0.1511	0.1581	0.1464
1200	0.1500	0.1496	0.1500	0.1502	0.1461	0.1595
1300	0.1500	0.1507	0.1502	0.1491	0.1506	0.1397
1400	0.1500	0.1497	0.1498	0.1504	0.1489	0.1522
1500	0.1500	0.1497	0.1502	0.1507	0.1465	0.1587
1600	0.1500	0.1508	0.1501	0.1491	0.1543	0.1500
1700	0.1500	0.1501	0.1496	0.1499	0.1496	0.1449
1800	0.1500	0.1487	0.1504	0.1511	0.1461	0.1463

Table 7.2: Example 1 of [11], recovering $a(y)$ at the strikes $K = 600$ to $K = 1800$ for the five test cases.

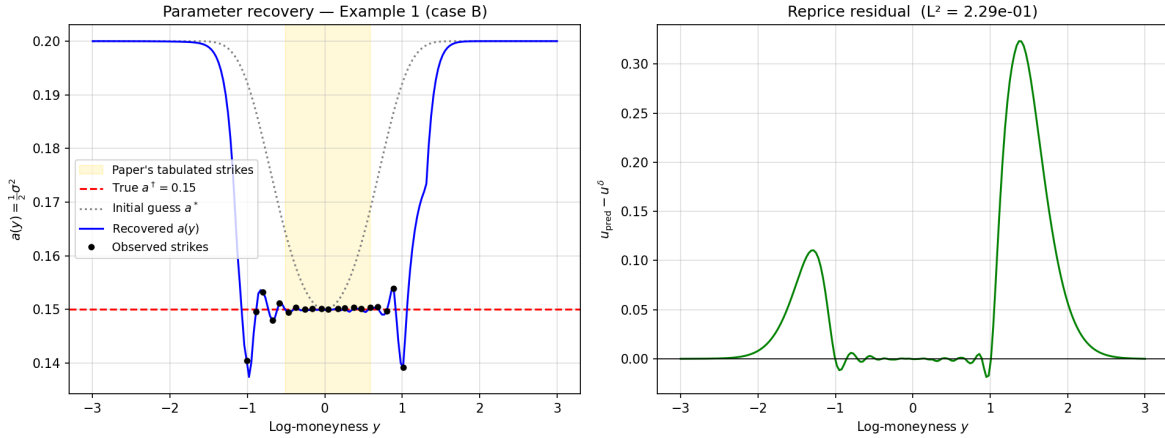


Figure 7.1: Example 1 case B of [11], recovering the constant $a(y) = 0.15$ using only 20 strikes.

7.10.1 Numerical experiment - Example 2

[11, Example 2] tries to show the importance of regularization, we implement it in `examples/example2.py`, see Appendix C. The set up is similar to example 1, but its true coefficient $a^\dagger(y)$ is the oscillatory profile near ATM

$$(7.18) \quad a^\dagger(y) = 0.15 + 0.05 e^{-(y+0.3)^2} \sin(2\pi y) + 0.05 \operatorname{erf}(20y).$$

To avoid an *inverse crime* we generate the synthetic data on a wider domain $[-5, 5]$, with a four-times finer mesh and time step, and interpolate them onto the coarser reconstruction grid, so the data do not come from the same discretized operator that is later inverted. We do it because [11] mentions to use a different time-integration scheme for the same reason, but does not mention

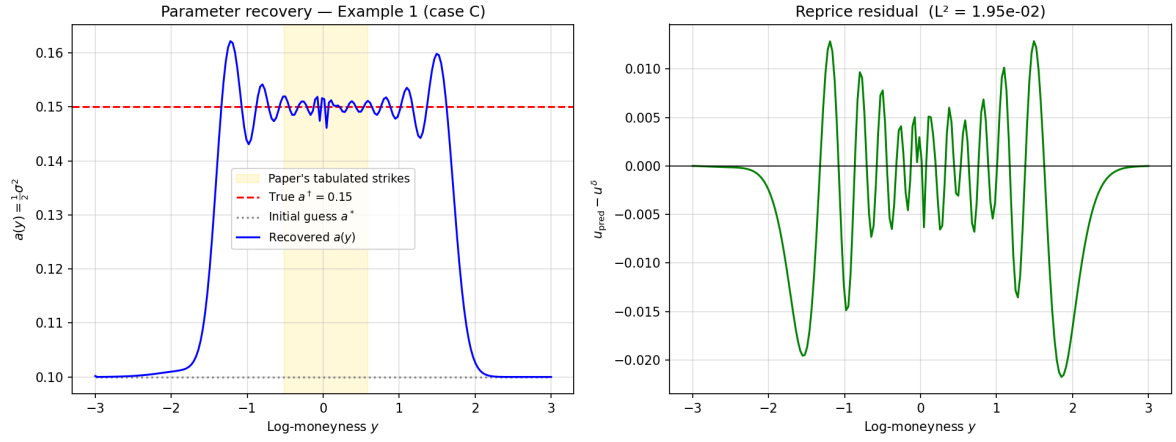


Figure 7.2: Example 1 case C of [11], recovering the constant $a(y) = 0.15$ from a poor constant prior $a^* = 0.10$.

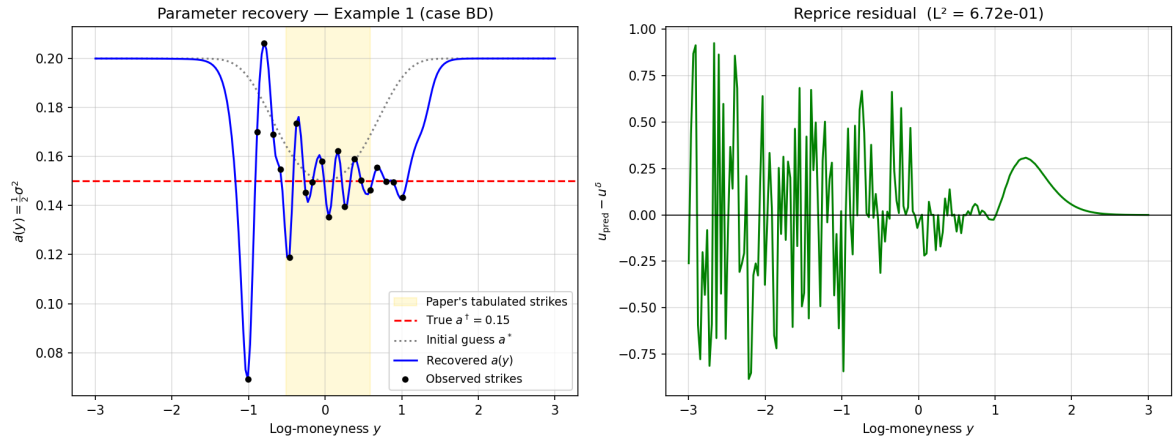


Figure 7.3: Example 1 case BD of [11], recovering the constant $a(y) = 0.15$ from incomplete and noisy data.

the details, in some of the cases looks like we are getting better reconstruction results and that might be the reason.

We run the paper's five cases. All use the *a priori* guess

$$a_1^*(y) = 0.15 + 0.05 \operatorname{erf}(2y),$$

which has the correct wings, except case C,

- (A) complete, exact data;
- (B) incomplete data, only 20 strikes observed;
- (C) complete, exact data with the poor constant prior $a^* = 0.15$;

(D) complete data with 0.1% noise;

(BD) incomplete and noisy.

For the exact data cases we use a single small $\beta = 10^{-6}$. Whereas for the noisy cases D and BD select β with the decreasing β schedule and Morozov discrepancy principle described above, starting from $\beta_0 = 10^2 \delta$ and warm-starting each minimization from the previous one.

Figure 7.4 shows the recovered coefficient for the five cases. Most exemplifying one is C, where we can see the importance of the prior, the poor constant prior $a^* = 0.15$ is not able to recover the oscillatory true value $a^\dagger(y)$, specially in the wings where the data is weak and the prior dominates the solution.

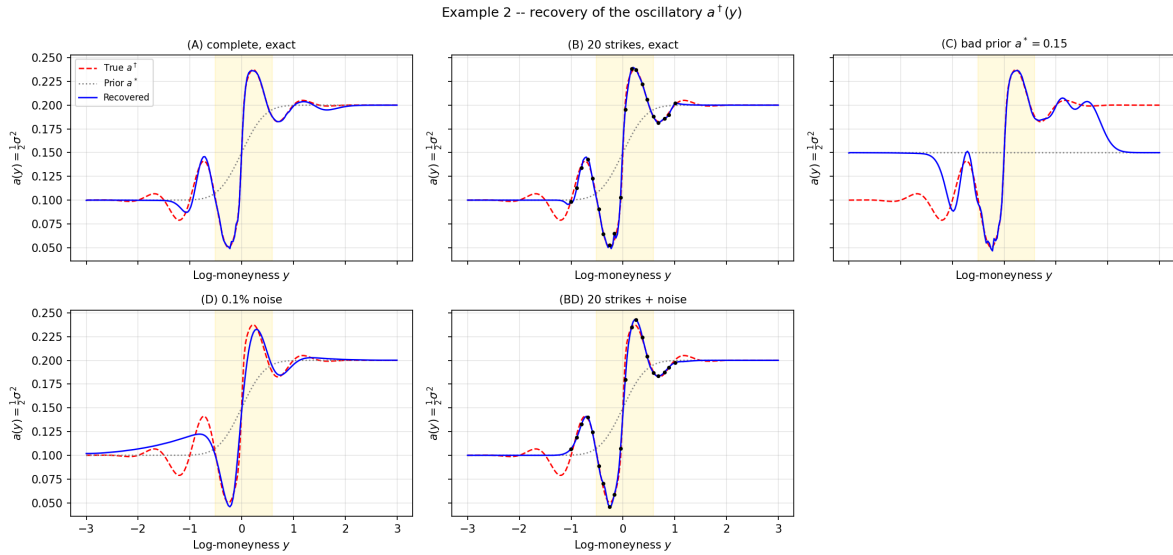


Figure 7.4: Example 2 of [11]: recovery of the oscillatory local variance $a^\dagger(y)$ (7.18) (red dashed) for the five test cases. Blue: recovered $a(y)$; grey dotted: prior a^* ; black dots (B, BD): observed strikes; shaded band: the strikes $K = 600$ – 1800 tabulated in the paper.

7.11 Numerical experiment - a local volatility surface from market data

Examples 1 and 2 use synthetic data and a time independent coefficient $a^\dagger$. We now calibrate a full surface $a(y, \tau)$ from real market quotes of the S&P 500 index, the same data we used for the implied volatility surface of Section 4.4.1. The rate theory of Section 7.6 assumes a time independent coefficient, so here we have no convergence rate, the surface fit is a numerical experiment. The script is `examples/example4_surface.py`, see Appendix C.

The data are SPY European option quotes from 30 April 2026, with spot $S_0 = 720.65$, on eleven expirations with maturities τ from 0.10 to 0.88 years. The rate, the dividend yield and

the smile of each expiration come from the pipeline of Appendix A. We keep the moneyness band $0.8 \leq K/S_0 \leq 1.2$, where the quotes are liquid, that is, quoted with a narrow bid-ask spread and traded in volume, so their mid price carries information rather than the width of the spread. Appendix B.8 lists the remaining filters.

The unknown is the stack of spatial slices $(a_1, \dots, a_K)$ of (7.16), one per maturity, solved for jointly by minimizing (7.17). The forward equation is marched once to the last maturity and read off at each one, so every quote samples the same evolving solution, and one forward solve with one adjoint sweep per iteration gives the whole gradient.

The prior a_k^* deserves care. Since $a = \frac{1}{2}\sigma^2$, the instinct is to seed slice k with the implied variance $\frac{1}{2}\sigma_{iv}^2$ of that maturity, which is at least in the right units. This is the wrong choice. The H^1 term penalizes the difference of slopes $\partial_y(a_k - a_k^*)$, and by the factor of two of Section 7.12 the implied skew is half the local one, so the reconstruction would be anchored at half the correct slope wherever the data are weak. We instead seed with the local volatility that the smile itself implies. Differentiating the short maturity relation (B.1) of [2] in y and solving for σ_{loc} gives

$$(7.19) \quad \sigma_{loc}(y) = \frac{\sigma_{iv}(y)}{1 - y \partial_y \sigma_{iv}(y) / \sigma_{iv}(y)},$$

which agrees with the smile at the money and doubles its skew there, as (7.20) of the next section requires. Appendix B.6 collects the details of the inversion.

Figures 7.6 to 7.8 show the result for $\beta_y = 0.1$ and $\beta_\tau = 1$. The surface has the shape one expects, a pronounced negative skew, with local volatility running from 0.15 at the money at the shortest maturity to 0.23 at the longest, near 0.12 on the upside, and rising on the downside to 0.40 at the long maturities and to 0.95 at the short end. Repricing the quotes gives a median relative error of 0.13%, with the maximum over liquid strikes below 1% at every expiration. The naive prior $\frac{1}{2}\sigma_{iv}^2$ gives a median of 0.20% and maxima above 2%, so (7.19) does remove the bias. As in the synthetic examples the surface is well determined near the money and on the downside, and relaxes towards the prior on the upside wing where call prices are close to zero, the ill-posedness of Section 7.6 seen on real data.

The calibrated surface is free of arbitrage by construction. The optimizer keeps the local variance strictly positive, $a \geq 10^{-8}$, and the prices solve the forward Dupire equation with that coefficient, so the output is the price surface of a genuine diffusion and the two conditions of Section 6.2.3 cannot be violated. The implied volatility route of Appendix A carries no such guarantee, there the smallest butterfly or calendar violation in the fitted surface feeds a negative local variance into the Dupire formula. Figure 7.5, produced by `examples/example7_arbitrage.py`, confirms it on the discrete surface, the risk-neutral density $\partial_{KK}^2 \mathcal{C}$ is non-negative across the observed band at every maturity, and the total implied variance $\sigma_{iv}^2 \tau$ increases in maturity at every moneyness.

Near the money the recovered local volatility is visibly steeper in strike than the market smile. This is a definite relation between the two skews, which we make precise next.

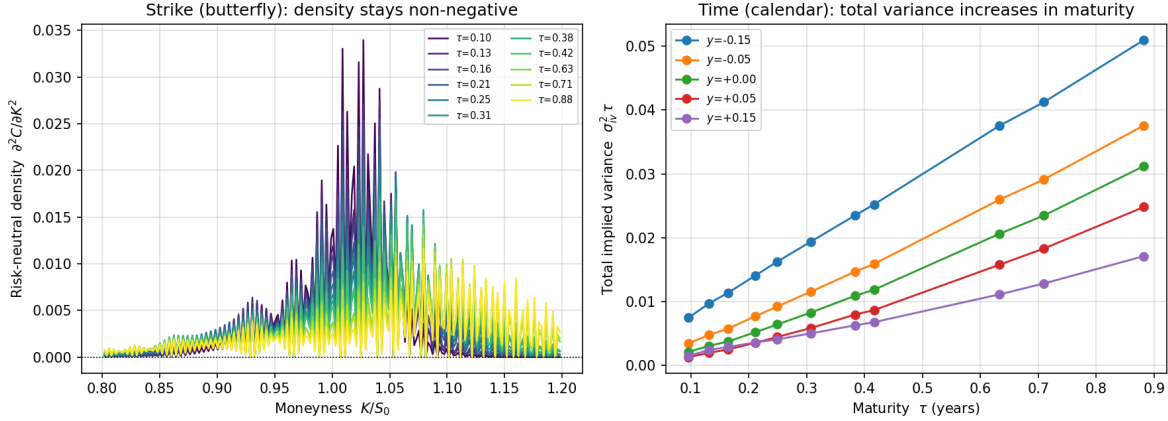


Figure 7.5: No-arbitrage check of the calibrated surface of Figure 7.6. Left: the risk-neutral density $\partial^2 \mathcal{C} / \partial K^2$ per maturity, non-negative across the observed band (no strike, or butterfly, arbitrage). Right: the total implied variance $\sigma_{iv}^2 \tau$ against maturity at fixed log-moneyness, increasing everywhere (no time, or calendar, arbitrage).

One feature of the fit points to the next section. Near the money the recovered local volatility is visibly steeper in strike than the market implied volatility plotted on the same axes. This is a definite relation between the two skews, which we make precise next following Derman and Bergomi.

7.12 The local volatility skew and the factor of two

In Figure 7.7 we can observe that local volatility has an steeper skew than the implied one. This is not a consequence of the calibration, it is the rule of two shown in Section 6.3.2.

Given the relationship between local volatility and implied volatility, from (6.12), near ATM the local volatility skew is twice the implied volatility skew, we can rewrite it as

$$(7.20) \quad \partial_y \sigma_{loc} \big|_{y=0} = 2 \partial_y \sigma_{iv} \big|_{y=0},$$

see [3] Chapter 2.

To verify such relationship numerically, we will check (7.20) directly on the calibrated surfaces, with `examples/example5_skew.py` of Appendix C. For each maturity τ , we fit a quadratic to the local volatility and to the smile in a narrow window $|y| \leq 0.05$ around the ATM and read the slope at $y = 0$. In figure 7.9 we can see two of the eleven maturities, drawing from the local volatility level a line with twice the implied slope lands on the local tangent, which is (7.20) read off the picture. Table 7.3 displays the two slopes at every maturity. The ratio averages 2.16 below three months, and stays between 2.0 and 2.5 over the whole term structure.

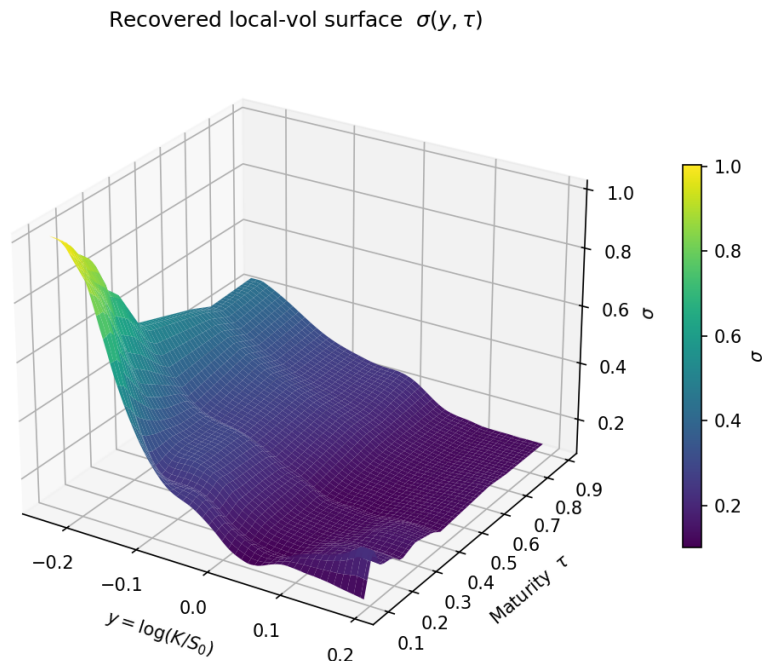


Figure 7.6: Local volatility surface $\sigma(y, \tau)$ calibrated from SPY quotes on 30 April 2026 ($S_0 = 720.65$, eleven maturities), with $\beta_y = 0.1$ and $\beta_\tau = 1$.

expiration	τ	$\partial_y \sigma_{iv}$	$\partial_y \sigma_{loc}$	ratio
2026-06-05	0.096	-0.827	-1.659	2.01
2026-06-18	0.132	-0.718	-1.558	2.17
2026-06-30	0.164	-0.676	-1.364	2.02
2026-07-17	0.211	-0.637	-1.504	2.36
2026-07-31	0.249	-0.611	-1.368	2.24
2026-08-21	0.307	-0.572	-1.328	2.32
2026-09-18	0.384	-0.518	-1.138	2.19
2026-09-30	0.416	-0.522	-1.171	2.24
2026-12-18	0.633	-0.452	-1.114	2.47
2027-01-15	0.710	-0.423	-0.919	2.17
2027-03-19	0.882	-0.390	-0.927	2.38

Table 7.3: At the money skews of the calibrated surface of Figure 7.6. Both slopes are the linear coefficient of a quadratic fit in $|y| \leq 0.05$, one on the market smile and one on the recovered local volatility slice, and the last column is their ratio, the factor of two of (6.12).

7.13 Out of sample test: pricing between quoted maturities

Both the calibrated local volatility surface and the interpolated implied volatility surface, agree on the price of quoted maturities by construction. But they differ on contracts that falls between

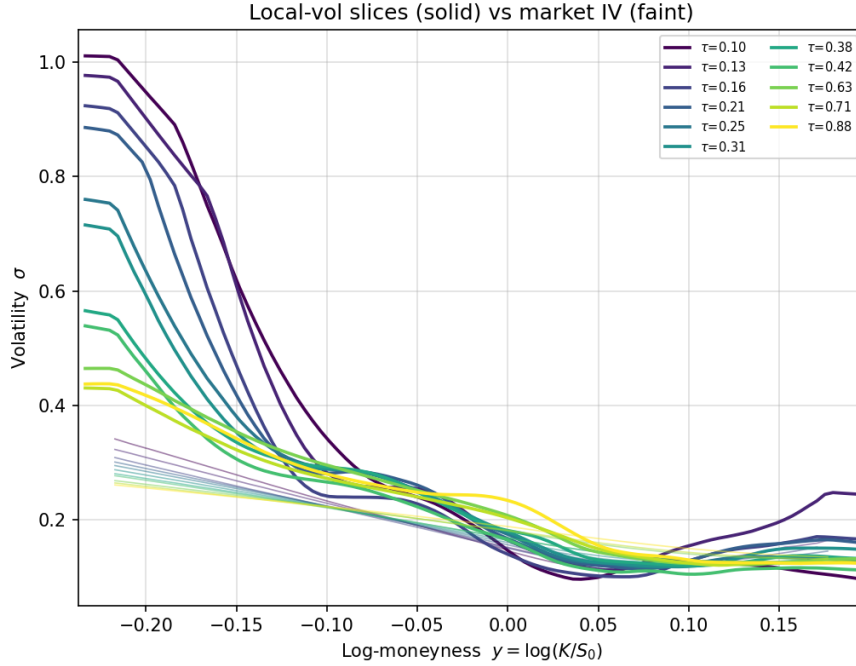


Figure 7.7: Recovered local volatility slices (solid) of the surface of Figure 7.6 against each maturity’s market implied volatility (faint); the local volatility skew is steeper than the implied one.

them.

To test how each method prices out of sample, in `examples/example6_holdout.py` of Appendix C, we entirely remove one interior expiration from the data, the December 2026 quotes at $\tau = 0.63$, and price it out of sample two ways. The local volatility surface is calibrated on the remaining ten maturities and the Dupire equation is used to recover the prices across the gap, so the price at $\tau = 0.63$ is a genuine sample of the forward solution rather than an interpolation of the smile. On the other hand, the implied volatility is computed at the remaining data points, then we interpolate the total variance $\sigma_{iv}^2 \tau$ linearly in maturity at fixed log-moneyness between the two closest expirations, $\tau = 0.42$ and $\tau = 0.71$, to finally calculate the price from Black-Scholes formula.

Figure 7.10 shows the outcome over the well quoted band $|y| \leq 0.15$. The deep ITM and OTM strikes are left out of the picture, deep ITM options are dominated by the intrinsic value $S_0 e^{-q\tau} - K e^{-r\tau}$, so they carry no volatility information, deep OTM are less liquid and the relative miss-price is higher due to low original price, the more far from the money an option is the cheaper it is. Both methods reproduce the excluded market prices closely, and neither is ever off by more than 2.0% over the band. Over the liquid strikes the median relative price error is 0.29% for the implied volatility interpolation and 0.75% for the local volatility surface. The interpolation is therefore the more accurate of the two, this is not a surprise. Interpolating the total variance is

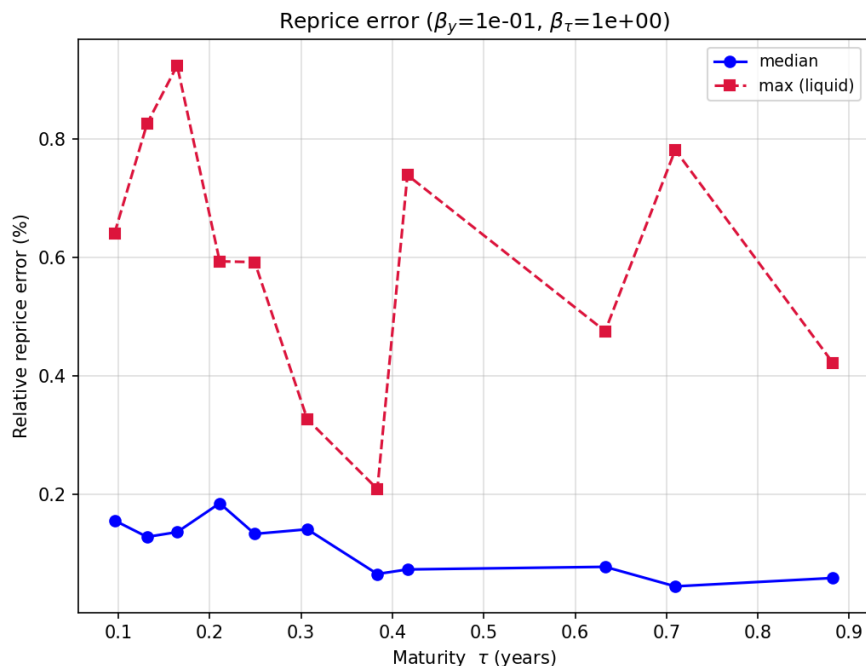


Figure 7.8: Per maturity relative repricing error of the calibrated surface of Figure 7.6 over liquid strikes (market price at least 0.5% of S_0), median and maximum.

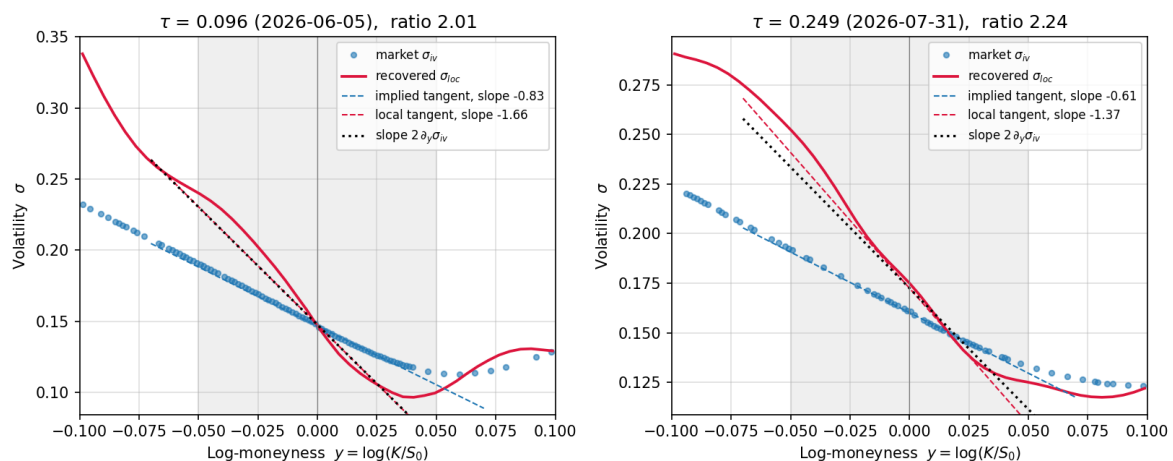


Figure 7.9: The rule of two on the SPY surface of Figure 7.6, at the shortest maturity (left) and at three months (right). Market implied volatility (dots) and recovered local volatility (solid) near the money, with the tangents at $y = 0$ of the quadratic fits over the shaded window $|y| \leq 0.05$. The dotted line has twice the implied slope and sits on the local tangent.

built to reproduce the smile, it uses the two nearest expirations to interpolate and the market volatility is exact at both of them, whereas the local volatility surface is a function of the *coefficient* of a partial differential equation, and it is propagated forward between the observed market

points, so an error in $a(y, \tau)$ accumulates in the solution before it is read off as a price.

Even though the implied volatility looks better on this numerical experiment, we should not conclude that is a better model, at the end of the day local volatility model is constructed under the not arbitrage premise, and market quotes are not always arbitrage free, thus reproducing the market prices is not necessarily a sign of a better model.

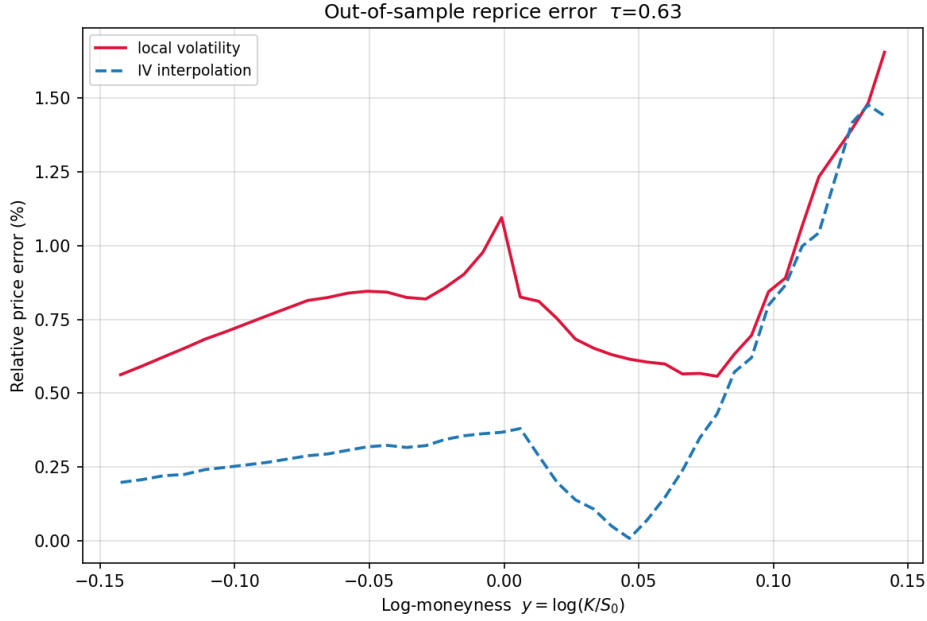


Figure 7.10: Out-of-sample pricing of a held-out expiration (December 2026, $\tau = 0.63$), removed from the calibration and bracketed by $\tau = 0.42$ and $\tau = 0.71$. Relative price error $|C - C_{\text{mkt}}|/C_{\text{mkt}}$ over the well quoted band $|y| \leq 0.15$, for the local volatility surface and for the implied volatility interpolation. Neither exceeds 2.5%, and the interpolation is the more accurate across the band.

CONCLUSIONS

This work has build up the mathematical financial theory of options up to the local volatility model, while presenting numerical results for the calibration of both, implied volatility surface and local volatility surface.

IMPLIED VOLATILITY SURFACE - IMPLEMENTATION DETAILS

This appendix describes the pipeline used to construct the SPY implied volatility surface shown in Section 4.2, including the practical difficulties encountered and the assumptions made at each step. All code lives in [Implied Volatility Code](#) and is reproducible from the cached market snapshot dated 2026-04-30.

A.1 Data

SPY option chains are fetched via the `yfinance` library. We restrict to maturities $4\text{ wk} \leq T \leq 1\text{ yr}$ and require a minimum trading volume of 10 contracts per quote. When both bid and ask are positive the mid-price is used; otherwise the last traded price is taken as a fallback (relevant for pre-market snapshots where the spread is stale but the last price is more informative). Data for calls and puts are cached separately as *parquet* files so that all subsequent computations are fully reproducible without network access.

A.2 Risk-free rate

The risk-free rate is taken from the US Treasury par yield curve published daily by the US Department of the Treasury. Par yields are available at standard maturities (1M, 2M, 3M, 4M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y) and are converted to continuously-compounded rates via

$$r_{\text{cc}} = \ln\left(1 + \frac{y}{100}\right),$$

where y is the par yield in percent. Rates are linearly interpolated (in T) between the published nodes and held flat outside the range. Figure A.1 shows the resulting curve on the reference date.

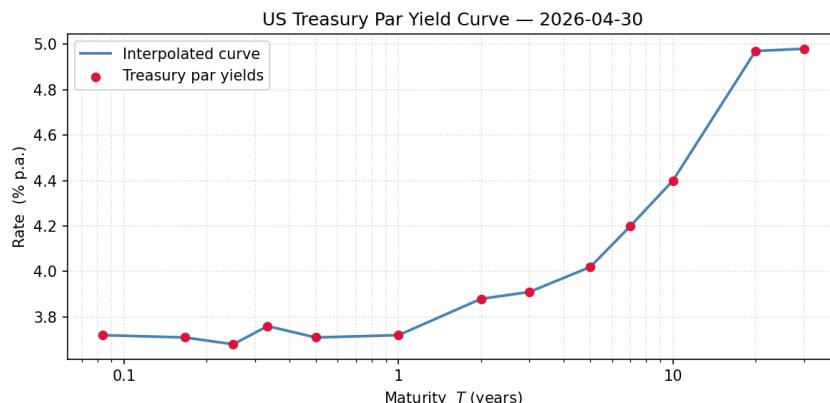


Figure A.1: US Treasury par yield curve on 2026-04-30 (log-maturity axis). Red dots are published nodes; the line is the linear interpolant used to assign a per-maturity discount rate.

Note that the risk-free rate could be refined further as the US Treasuries are not strictly risk-free, as they carry a small credit risk premium. A better discounting rate for SPY options would be the overnight index swap (OIS) rate rather than the Treasury curve, but for our purposes the difference is not enough to justify the additional complexity of fetching and processing OIS data.

A.3 Dividend yield calibration

SPY pays dividends quarterly, so any pricing formula must account for the dividend yield q . Getting q wrong breaks the internal consistency of the surface: puts and calls at the same strike and maturity imply different volatilities, producing a visible discontinuity at the money in any composite surface. We went through three successive approximations before reaching a satisfactory result.

Stage 1 - $q = 0$. The simplest assumption is to ignore dividends entirely. This is equivalent to treating SPY as a non-dividend-paying asset, which is a standard simplification when the dividend yield is small relative to the interest rate and the maturity is short. For maturities beyond a few months, however, cumulative dividends are non-negligible (SPY's trailing yield is approximately 1.1%) and the forward price implied by put-call parity,

$$F = S e^{(r-q)T},$$

is overestimated if q is set to zero. As a result, the Black-Scholes-Merton formula assigns different implied volatilities to calls and puts at the same strike, since each type is sensitive to the forward in a different direction.

Stage 2 - Fixed annual yield. The natural correction is to use the trailing twelve-month dividend yield reported by the data provider ($q \approx 1.14\%$ for SPY on the reference date). This

constant is applied uniformly to all maturities. The surface improves near the money, but the fix is imprecise: SPY dividends are paid on a quarterly schedule, so the effective yield between two option expirations depends on how many ex-dividend dates fall within that window. A short-dated option expiring before the next ex-dividend date carries an effective $q \approx 0$, while a longer-dated option spanning several dividend payments accumulates a yield closer to the annual figure. Applying a single annual constant systematically over-corrects for short maturities and under-corrects in an uncontrolled way across the term structure.

Stage 3 - Put-Call parity calibration. The cleanest approach is to infer q at each expiration directly from market prices using put-call parity, keeping the Treasury rate r fixed. For each matched call-put pair at strike K_i and expiration T , put-call parity gives

$$C_i - P_i = S e^{-qT} - K_i e^{-rT}.$$

Defining the cost-of-carry forward $F = S e^{(r-q)T}$ and multiplying both sides by e^{rT} yields an implied forward per pair,

$$F_i = (C_i - P_i) e^{rT} + K_i,$$

where q is encoded inside F_i rather than appearing explicitly. Taking the median over the relevant pairs at a given expiration yields a robust forward estimate $\tilde{F}$, and inverting $\tilde{F} = S e^{(r-q)T}$ gives

$$q = r - \frac{\ln(\tilde{F}/S)}{T}.$$

A key implementation choice is *which pairs to include*. Put-call parity holds model-free for any European options, but the quality of the forward estimate depends on how cleanly each leg is priced. For a strike $K < S$ the call is in-the-money and the put is out-of-the-money; for $K > S$ the roles are reversed. Deep in-the-money options are significantly less liquid: their price is dominated by intrinsic value, the bid-ask spread is wide relative to the time value, and last-trade prices can be hours stale. Feeding these pairs into the forward estimate biases $\tilde{F}$ toward noisy values.

We therefore restrict the calibration to near-ATM pairs satisfying $|K/S - 1| \leq 0.05$ (i.e. strikes within $\pm 5\%$ of the current spot). In this band, both the call and the put are close to at-the-money; both legs have tight spreads and meaningful open interest, and the time value is large relative to the bid-ask noise. The median over these pairs (rather than the mean or a least-squares fit) guards against the occasional stale quote even in the near-ATM region. If fewer than two near-ATM pairs are available the constraint is relaxed to all matched pairs.

Since r is fixed from the Treasury curve, there is only one free parameter per expiration, which avoids the instability of simultaneously fitting both r and q . Calibrated values are bounded to $q \in [-0.02, 0.10]$; if the result is outside this range (a sign of too few matched pairs or erroneous data), the provider value is used as a fallback.

The resulting term structure shows $q \approx 0$ for near-term expirations (the next quarterly dividend has not yet been declared within the window) and a gradual convergence to the trailing annual yield for longer maturities, which is economically consistent with the quarterly payment schedule.

Figure A.2 illustrates the three stages for the December 2026 expiration ($T \approx 0.63$ yr), where the effect of the dividend yield is large enough to be clearly visible. Each panel overlays the put-implied and call-implied volatility curves across the full strike range. With $q = 0$ (Stage 1) the two curves diverge significantly and the level is shifted upward for OTM puts and downward for OTM calls. With a fixed annual q (Stage 2) the ATM region improves but the term-structure mismatch remains. With the PCP-calibrated q (Stage 3) the two curves sit on top of each other near the money, confirming that the composite surface can be joined smoothly. The residual divergence at deep in-the-money strikes visible in all three panels is due to liquidity issues (wide bid-ask spreads, stale prices) rather than the dividend yield assumption, and is removed by restricting the composite to out-of-the-money quotes only.

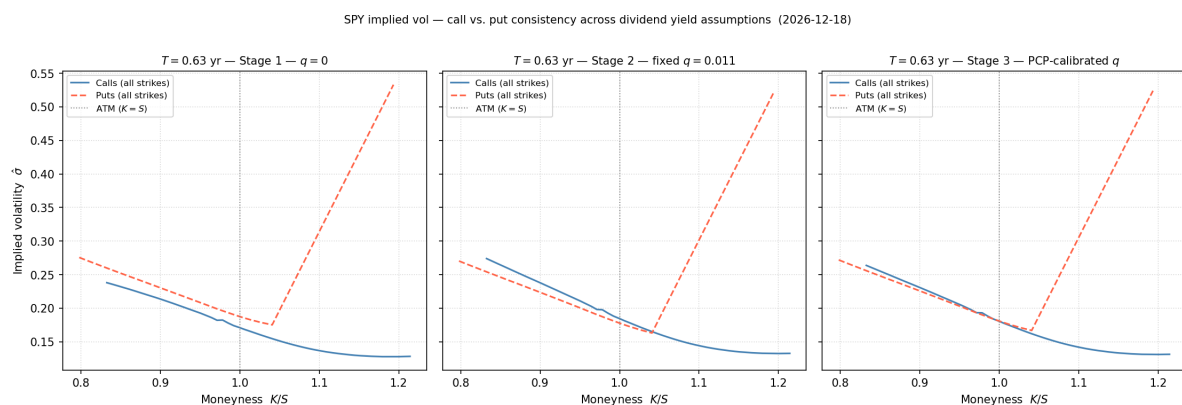


Figure A.2: Call-implied (solid blue) and put-implied (dashed red) volatility across all available strikes at the December 2026 expiration ($T \approx 0.63$ yr), under three dividend yield assumptions. Stage 1 ($q = 0$): curves diverge sharply — the forward is overestimated, shifting call and put IVs in opposite directions. Stage 2 (fixed annual q): improved near ATM, but the flat term structure leaves systematic errors for maturities where the true q differs from the annual average. Stage 3 (PCP-calibrated q from near-ATM pairs): the two curves coincide in the OTM region, where both legs are liquid. The residual divergence far from ATM (deep ITM options) is a liquidity artefact — those strikes are excluded from the composite surface.

A.4 OTM composite construction

Once r and q are consistent with market prices, the composite surface is formed by taking out-of-the-money options only: OTM puts ($K < S$) for the left wing and OTM calls ($K \geq S$) for the right wing.

The rationale is that in-the-money options are significantly less liquid than their out-of-the-money counterparts: ITM bids tend to be stale, spreads are wide, and the option price is dominated by intrinsic value, so a small absolute pricing error translates into a large relative error in the time value — and therefore a large error in the inverted implied volatility. This is visible in Figure A.2: the divergence between call and put IV curves away from the money persists even in Stage 3, not because q is wrong but because deep-ITM prices are unreliable.

Figures A.3 and A.4 show the full call and put surfaces (all strikes, no ITM filter) to illustrate the noise. The clean composite (Figure 4.3) is obtained after restricting to $0.75 \leq K/S \leq 1.25$ and keeping only the OTM leg for each option type.

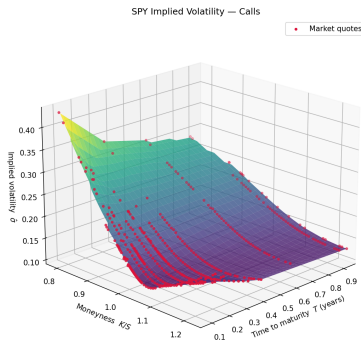


Figure A.3: IV surface from all call quotes (including deep ITM, $K \ll S$). Deep ITM calls carry noisy implied volatilities due to wide spreads and stale prices.

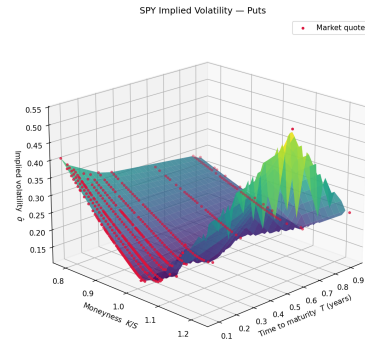


Figure A.4: IV surface from all put quotes (including deep ITM, $K \gg S$). Same liquidity issue for strikes far above the spot.

A.5 Expiration selection

SPY has a dense cluster of weekly expirations in the short end (roughly 1-8 weeks). Including all available expirations would over-weight this region and produce a surface that appears smooth in absolute maturity but poorly samples the longer end. To obtain a balanced representation, the full maturity range is divided into $N = 10$ log-spaced bins; within each bin the expiration with the highest combined open interest (calls plus puts) is kept. This ensures that liquidity guides the selection and that the resulting sample is approximately uniform in $\log T$, which is the natural scale on which the term structure of volatility varies.

A.6 Summary of assumptions

For reference, the key modelling assumptions are:

- **European exercise:** SPY options are technically American-style. The early-exercise premium is small for an index ETF with a low continuous yield and is neglected here; the BSM formula is used as an approximation.
- **Continuous dividends:** actual SPY dividends are discrete quarterly payments. The PCP calibration implicitly converts them into a continuous yield that is consistent with the current option prices, which is the standard market convention.
- **Moneyness filter** $0.75 \leq K/S \leq 1.25$: strikes outside this range are too illiquid to contribute reliable IV observations.
- **Volume filter:** quotes with fewer than 10 contracts of daily volume are excluded.
- **Single snapshot:** the surface is computed from a single end-of-day snapshot; intraday bid-ask movements are not captured.

LOCAL VOLATILITY SURFACE - IMPLEMENTATION DETAILS

This appendix collects the implementation of the local volatility calibration of Chapter 7 and the decisions taken along the way that the main text states without justifying. All code lives in **Local Volatility Code** and runs from the same cached SPY snapshot dated 2026-04-30 used throughout Appendix A. The first sections describe the discrete problem that is actually solved, the domain, the parametrization of the surface, the observation operator and the gradient sweep. The later ones record the choices that could reasonably have been made differently, the prior, the term structure, the selection of expirations, the regularization parameters and the normalization of the unknown.

B.1 Discretization and computational domain

The discretization is the one announced in Section 7.8, P_1 elements in y and backward Euler in τ . What the main text leaves open is the size of everything.

The spatial domain is not fixed in advance, it is read off the data. Each kept expiration contributes the log-moneyness range of its quoted strikes inside the band $0.8 \leq K/S_0 \leq 1.2$, the union of those ranges is $y \in [-0.217, 0.177]$, and the computational domain is that union padded by 0.25 on each side, $y \in [-0.467, 0.427]$, that is strikes from $0.63S_0$ to $1.53S_0$. The pad is what separates the artificial boundary from the data. Its width is about 1.8 standard deviations of the longest maturity, $\sigma\sqrt{\tau} \approx 0.14$ at $\sigma = 0.15$ and $\tau = 0.88$, so whatever error we commit at the boundary is damped by the diffusion before it reaches the observed band. Widening it further only buys solve time, since the coefficient out there is constrained by no quote at all.

The mesh is uniform, 199 elements and therefore 200 nodes, a spacing of $h \approx 4.5 \times 10^{-3}$ in y , under half a percent in strike and finer than the quoted strike spacing everywhere in the band.

Time is divided into 300 steps up to the last maturity, $\Delta\tau \approx 2.9 \times 10^{-3}$ years, close to one calendar day.

Backward Euler is a deliberate choice over Crank-Nicolson. Crank-Nicolson is second order in time, but it is not strongly damping: its amplification factor tends to -1 for the stiff modes, so a non-smooth initial condition leaves oscillations that decay only slowly. Our initial condition is the call payoff, whose kink at $y = 0$ excites exactly those modes, and the oscillations are worst in the second derivative in strike, which is the quantity carrying the information about α . The standard cure is a Rannacher startup of a few implicit steps [14]. We use implicit Euler throughout instead, which is uniformly damping at first order, because the time error is not the binding constraint here: Table B.1 below measures it at a day-sized step as 5×10^{-2} dollars on a spot of 720, a relative 10^{-4} , an order of magnitude below the reprice errors we report.

The solver works in dollars, not in the normalized price $u = C/S_0$ of Section 7.2. The equation is linear so the two differ by a constant factor and describe the same coefficient, but the data term scales as S_0^2 while $\beta \| \alpha - \alpha^* \|_{H^1}^2$ does not, so β is tied to the scale of the prices. The values quoted below are for prices in dollars at a spot of 720, and they cannot be carried to another underlying, or to the normalized form, without rescaling.

The boundary conditions are Dirichlet at both edges. The limits recorded in (7.3), intrinsic value on the left and zero on the right, hold as $M \rightarrow \infty$ and are too crude at a pad of 0.25, where a call struck at $0.63S_0$ is still worth appreciably more than its intrinsic value at long maturity. We instead evaluate the Black-Scholes price at the two edge strikes, using that maturity's zero rate and dividend yield and a wing volatility interpolated across the term structure from the extreme quoted strike of each smile. This is exact whenever the smile is flat outside the quoted band, which is the same assumption already built into the flat extension of the prior. At $\tau = 0$ the boundary value is the payoff, so the data are continuous at the corner.

B.2 The surface as a stack of time slices

The unknown is the stack of spatial slices of (7.16), one per quoted maturity. With eleven maturities and 200 nodes that is 2200 numbers, held as a single flat vector, which is what the optimizer sees and what the bound constraints act on.

The slice boundaries have to live on the time grid, so each maturity is snapped to the nearest time level, and if two maturities land on the same level the later one is pushed forward by one step. On this data the eleven maturities map to distinct levels and the longest lands exactly on the final step. The maturity the model prices is therefore displaced from the quoted one by at most half a step, about half a day, and we do not correct for it: the displacement is well inside the uncertainty of an end of day quote.

Slice k then covers $(\tau_{k-1}, \tau_k]$ and the observation at τ_k is read off the trajectory at exactly the level where slice k ends. This is what makes the triangular structure of Section 7.8.2 exact in the

discrete problem: the first expiration sees only the first slice, the second one the first two, and so on.

The optimizer imposes $a \geq 10^{-8}$ and no upper bound, although the admissible set (7.4) carries one. The lower bound is the one that matters, it keeps the operator parabolic and the recovered volatility real, and it is what makes the calibrated surface arbitrage-free by construction.

B.3 The residual, and how a smile becomes data

The misfit is the finite element L^2 norm over the observed band, not a sum of squared errors over the quotes. Each expiration's call prices are interpolated linearly onto the mesh nodes, and the difference against the computed price is multiplied by the $\{0, 1\}$ indicator of the nodes lying inside that expiration's quoted range. Because the mask is a projection, $P^2 = P$, the same masked residual is simultaneously the integrand of the data term and the source injected into the adjoint sweep, and no separate transpose has to be worked out.

The consequence of measuring the misfit in a continuum norm is that quotes are weighted by mesh spacing rather than by how many of them there are. The expiration with 116 strikes in the band and the one with 46 contribute the same integrated weight over the same interval of moneyness, and a region where the chain quotes every dollar does not outweigh one where it quotes every five. A plain sum over quotes would tilt the fit toward the densely quoted expirations and toward the money, for no modelling reason.

The mask is a hard cut at the ends of each quoted range. Nothing outside is fitted, so the coefficient in the wings is set by the penalty alone, which is exactly the behaviour visible in the recovered slices of Figure 7.7.

B.4 The gradient in a single backward sweep

One evaluation of the functional and its gradient is the following sequence.

- (i) March the forward equation from $\tau = 0$ to the last maturity, storing the whole trajectory. Here that is 301 states of 200 values, so everything fits in memory and no checkpointing is needed.
- (ii) At each observed level, form the masked residual, accumulate its squared norm into the data term, and set the residual aside as an injection.
- (iii) March the adjoint equation backward from the last level, started from the last maturity's residual.
- (iv) At each step, after solving for the new adjoint state, assemble that step's contribution to the gradient and add it to the slice that owns the step.

(v) Only then, add the residual observed at that level to the adjoint state.

(vi) Add the gradient of the penalty.

The order of (iv) and (v) is the part that is easy to get wrong. Injecting the residual of τ_j before taking the step's gradient contribution would let the price quoted at τ_j feed the coefficient on $(\tau_j, \tau_{j+1}]$, that is, it would make an option depend on the volatility after its own expiry. Taking the contribution first keeps the dependence causal and reproduces the triangular structure at the discrete level.

Two further points deserve recording.

The gradient is assembled directly as a linear form against the test functions and stopped there, with no mass matrix solve. Solving $Mg = b$ would instead return the Riesz representative of the derivative in L^2 , which is the natural gradient of the continuous problem, whereas the assembled vector b is the derivative with respect to the nodal coefficients. Both are legitimately called the gradient, in different inner products, and mixing the two is the usual way to obtain a gradient test that passes on one mesh and fails on the next. We use the coefficient version everywhere, and the penalty gradient is assembled to match, as $2\beta(M + K)(a - a^*)$ with M and K the mass and stiffness matrices, so the two terms of the functional are differentiated in the same sense. It is also the version L-BFGS-B expects, since its line search and its box constraints act on the coefficients.

What is differentiated is the discrete forward operator, not the continuous one. Every term of the variational form (7.13) carrying a contributes: the diffusion, the $\partial_y a$ term produced by the integration by parts, and the au_y half of the convection. This is why the duality test of Section 7.9.1 closes to rounding error, 2.3×10^{-15} , rather than to discretization error. Discretizing the continuous adjoint instead would have given a gradient that is correct in the limit and slightly wrong on the grid, which is enough to stall a quasi-Newton method near the minimum.

The cost is two PDE solves per iteration whatever the number of slices and the number of nodes. Calibrating the eleven slices of the surface costs the same per iteration as calibrating the single curve of Example 1.

B.5 The discrete penalty

The two penalties of (7.17) are assembled from the same matrices. The H^1 term is the full norm, values and slopes, and no condition is imposed on a at the edges of the domain, so the minimizer meets the boundary with a flat coefficient, which is consistent with the flat wings of the prior.

The penalty across maturities is an L^2 jump rather than a discrete derivative. The coefficient is piecewise constant in τ by construction, so $\partial_\tau a$ does not exist and $\|a_k - a_{k-1}\|_{L^2}^2$ is the natural discrete stand-in. It is deliberately not divided by the gap $\tau_k - \tau_{k-1}$. The gaps in this chain run from twelve days to seventy-nine, and dividing by them would let the surface move almost freely

across the long gaps, which is precisely where the data are thinnest and the smoothing is most needed.

Only the H^1 term sees the prior. The jump term is prior free, so it is a genuine smoothness requirement in maturity rather than a second pull toward the seed.

B.6 The prior and the seed

Section 7.11 explains why the prior cannot be the implied variance $\frac{1}{2}\sigma_{iv}^2$ and states the inversion (7.19) used instead. It comes from the short maturity result of [2], that the implied volatility is the harmonic mean of the local volatility between spot and strike,

$$(B.1) \quad \frac{y}{\sigma_{iv}(y)} = \int_0^y \frac{dx}{\sigma_{loc}(x)} \quad \text{as } \tau \rightarrow 0,$$

whose derivative in y gives (7.19) directly.

Two details of the inversion matter in practice. The smile is replaced by a quadratic fit over the band before it is inverted, because the inversion needs $\partial_y \sigma_{iv}$ and differentiating a raw quoted smile amplifies quote noise into the prior; a quadratic is the lowest order carrying both a skew and a curvature, so the seed keeps the two features that matter without importing the wiggles of one day's chain. And the denominator of (7.19) is floored at 0.35. On the downside wing the skew is steep enough for $1 - y \partial_y \sigma_{iv} / \sigma_{iv}$ to collapse toward zero, and a short maturity asymptotic used at τ close to a year has no business producing an unbounded seed; the floor caps it at roughly three times the implied level. Outside the quoted band the seed is extended flat.

The same field is used as the prior a^* and as the starting iterate, so the optimizer begins at the centre of the penalty and moves away from it only where the prices ask it to.

B.7 Interest and dividend as piecewise constant forwards

The implied volatility pipeline delivers one zero rate r_j and one dividend yield q_j per expiration. The forward equation needs an instantaneous rate at every time level, so both are bootstrapped into forwards that are constant on each interval between quoted maturities and chosen so that integrating up to τ_j returns $z_j \tau_j$. Each maturity is then priced with exactly the discount factor that was used to invert its smile, and the pair $r - q$ is Egger's time dependent drift. The boundary values are the one place where the zero rates are used rather than the forwards, since those are Black-Scholes prices for a maturity and not local coefficients.

Bootstrapping differentiates the curve, so it amplifies whatever noise the zero rates carry. For r this is harmless, the forwards stay between 3.5% and 3.9%. For q it is visible: the per-expiration put-call parity estimates are noisy at the tenth of a percent level, and the implied forwards swing between -0.4% and 3.6% , one interval slightly negative. We leave them as the bootstrap produces them. Smoothing the dividend term structure would be defensible, but the alternative of a single

constant q misprices every maturity's forward by construction, and the per-interval wobble enters only through the drift $r - q$ over one interval at a time.

B.8 From the option chain to the smiles

Everything upstream of the calibration is the pipeline of Appendix A, the same snapshot, the same Treasury rates, the same put-call parity dividend yields and the same inverted volatilities.

The prices handed to the calibration are call prices, since that is what the Dupire forward equation solves for, but they are not the quoted calls. Below spot the quoted put is converted by put-call parity, $C = P + S_0 e^{-q\tau} - K e^{-r\tau}$, so every price comes from the liquid out of the money leg while still describing a call. This is the out of the money composite of Appendix A re-expressed in call prices.

The selection of expirations is the one place where the two pipelines deliberately part. Appendix A keeps the most liquid expiration in each of ten log-spaced maturity buckets, which balances a surface that is only going to be drawn. Here the slices are coupled in time, so skipping a liquid expiration is not neutral: it leaves the calibration to cross a gap the market does not have, with the jump penalty rather than quotes deciding what happens inside it. The buckets also widen at the long end, exactly where the chain is already sparse. We therefore keep every expiration whose composite carries at least twenty five strikes inside the band, and drop the two that sit within two weeks of a kept one, 31 August and 31 December 2026. Two expirations a few days apart carry nearly the same information, and the second slice would be fixed by the penalty rather than by its own quotes. Eleven expirations survive, with maturities from 0.096 to 0.882 years and between 46 and 116 strikes each in the band.

B.9 Choice of the regularization parameters on market data

Section 7.7 describes Egger's decreasing sequence of β stopped by the Morozov discrepancy principle, which is what the synthetic examples use, where δ is known because the noise was added by hand.

On market data δ is not known. We estimated it from a relative price noise model, half a percent of each quoted price over the observed band, which is the order of a bid-ask spread on liquid SPY options, and ran the schedule against it. The stopping rule is not trustworthy at that point. The estimate is a guess rather than a measurement, and more importantly the gap between the quotes and the model is not noise in the sense the principle assumes: it also contains the early exercise premium neglected in Appendix A, the fact that the quotes across an expiration are not simultaneous, and the error in r and q . A stopping β inherits all of it, and moves with an estimate we cannot pin down.

We therefore fix β by hand for the market runs, $\beta = 10^{-4}$ for the single maturity fit and $\beta_y = 10^{-1}$, $\beta_\tau = 1$ for the surface, chosen by inspecting the trade-off directly: small enough that

the reprice error settles near the level of the bid-ask spread, large enough that the slices do not develop structure the quotes cannot support. As noted above, these numbers are attached to prices in dollars at a spot of 720 and to this mesh, and they are not dimensionless.

B.10 Reporting the reprice error

The reprice errors of Figure 7.8 are relative errors on prices, which are only meaningful where the price is not almost zero. We therefore report them over the liquid strikes of each maturity, those whose market price is at least 0.5% of spot, about \$3.60. Without that floor the far out of the money calls, worth a few cents, would dominate the maximum and say nothing about the quality of the fit. The model price at a quoted strike is read back by interpolating the computed solution from the mesh nodes. The unweighted L^2 misfit over the whole band is reported separately, so the two measures are not confused: they do not have to move together, and in Table B.2 they do not.

B.11 Decisions in the synthetic examples

Three choices in the reproduction of Examples 1 and 2 of [11] are worth recording, since none of them is stated in the paper.

The spot. The text of the paper says $S = 100$, but its tabulated option values, a call worth 439 at $K = 600$, and the description of test D, an underlying worth 1000\$, only make sense at $S_0 = 1000$. The recovered α is invariant under a rescaling of the prices, but the Tikhonov balance is not, the data term scales as S_0^2 and the penalty does not, so reproducing the paper's β requires its S_0 . We use $S_0 = 1000$.

The noise model. The paper adds 0.1% noise to the data and remarks that at $S_0 = 1000$ this amounts to an uncertainty of about $\pm 1\$$, of the size of a typical bid-ask spread. The two readings are not the same: 0.1% of S_0 is a flat $\pm 1\$$ band on every price, 0.1% of each price is proportional to it. We use the relative reading. A flat band is a far larger perturbation on the cheap out of the money calls, where the price itself is of the order of a dollar and the whole signal is destroyed, and it did not reproduce the paper's reconstructions; the relative reading does.

The inverse crime. The paper generates its synthetic data on a wider domain with a finer discretization and a different time integrator, so that the operator being inverted is not the operator that produced the data. We follow the same idea, generating the data of Example 2 on $[-5, 5]$ with four times the spatial and temporal resolution and interpolating onto the reconstruction grid, but keep implicit Euler on both sides rather than switching schemes. The refinement alone already separates the two operators.

B.12 The discounted normalization $u = e^{\int_t^T q ds} C$

Throughout Section 7.11 the unknown is the call price itself, in dollars (Section B.1), and the equation keeps the explicit zero-order term of (7.1). [11] instead absorb the dividend factor into the unknown by setting

$$\tilde{u}(y, \tau) = e^{\int_t^T q(s) ds} u(y, \tau),$$

which cancels the $-q u$ term and leaves the shorter equation

$$-\tilde{u}_\tau + a(\tilde{u}_{yy} - \tilde{u}_y) + (q - r)\tilde{u}_y = 0.$$

A deterministic factor relates the two normalizations, so they describe the same continuous problem and coincide exactly when $q = 0$, as in Examples 1 and 2. On the market surface q comes from put-call parity (Appendix A.3) and reaches about 1.1%, so the only thing that can separate them is the discretization. Which one discretizes better is not obvious. The discounted form treats the exponential decay in τ exactly instead of leaving it to the backward Euler scheme, but it pays for that by rescaling the data and the boundary values, which reweights the residual across maturities. We implemented both and compared them. Both remain in the code as a switch on the solver, and the two tables below are reproducible from it.

The discounted form needs three changes: drop the zero-order coefficient from the variational forms (7.13) and (7.14), multiply the observed prices and the Dirichlet data by $e^{\int_0^\tau q}$, and divide that factor out again when the calibrated surface is repriced. The two solvers do agree on the continuous problem. Holding a fixed at the implied volatility seed and converting the discounted solution back to prices, the two forward solves match to 9×10^{-3} dollars on a spot of 720. That is a relative difference of 10^{-4} , the $O(\Delta\tau)$ gap we expect.

Refining the time grid tells us which one is closer to the exact solution. We solved forward at the production grid of $N = 300$ time steps and compared against a reference computed on the same time levels with $N = 2400$. Table B.1 reports the error averaged over the eleven quoted maturities, and repeats the experiment at artificially inflated dividend yields.

Even at a dividend yield of 20%, an order of magnitude above anything SPY pays, the gain is a few percent. The same holds for the inverse problem. Running the full surface calibration (7.17) both ways gives surfaces that differ by up to one volatility point, which looks large until the two effects are separated. Table B.2 reprices each calibrated surface with both solvers. Swapping the solver moves the median error by 0.003 percentage points, swapping the surface moves it by about 0.01. The gap therefore measures the path the optimizer took through a flat valley of the Tikhonov functional, not the normalization. Neither surface is better than the other either, since the one with the lower median error over the liquid strikes carries the higher unweighted data misfit.

Neither approach outperforms the other on this data, so we keep the explicit zero-order form as the default. It compares the solver output directly against the quoted prices, with no rescaling

q	explicit $-q u$	discounted	ratio
market ($\leq 1.6\%$)	5.51×10^{-2}	5.38×10^{-2}	1.02
2%	5.25×10^{-2}	5.22×10^{-2}	1.01
5%	4.44×10^{-2}	4.37×10^{-2}	1.01
10%	3.37×10^{-2}	3.28×10^{-2}	1.03
20%	4.93×10^{-2}	4.68×10^{-2}	1.05

Table B.1: Time discretization error of the forward solve at $N = 300$ time steps, in dollars against an $N = 2400$ reference on the same time levels, averaged over the eleven quoted maturities. The first row uses the calibrated dividend term structure, the others a flat override. The discounted form never gains more than 5%, because the backward Euler error is dominated by the kink of the payoff and by the convection term rather than by the zero-order term.

calibrated with	repriced with	median error	data misfit (\$ ²)
discounted	discounted	0.116%	0.02881
discounted	explicit $-q u$	0.119%	0.02928
explicit $-q u$	discounted	0.131%	0.02776
explicit $-q u$	explicit $-q u$	0.128%	0.02817

Table B.2: Each calibrated surface of Section 7.11 repriced with both solvers ($\beta_y = 10^{-1}$, $\beta_\tau = 1$, eleven maturities). The median error is taken over the liquid strikes of each maturity, the data misfit is the unweighted term of (7.17). The two rows of each block differ only by the solver and barely move, so what separates the blocks is the optimizer path.

in between. The discounted form is the better choice on an underlying with a large carry, a high dividend single stock or an FX option, where the foreign rate plays the role of q and the factor $e^{\int_t^T q}$ is no longer close to one.



SOURCE CODE

Every figure, table and number reported in this work is reproducible from the repository github.com/M315/TFM. This appendix gives the layout of the code and says which script produces what, so that any result in the main text can be traced back to the routine that computed it. Appendices A and B discuss the choices made inside these routines, here we only describe the structure.

The code is organized in two packages under `scripts/`. The first builds the implied volatility surface from the raw option chain, the second calibrates the local volatility surface from the prices that the first exports. They run in separate conda environments, because the finite element solver uses a legacy FEniCS stack without pandas, which is why the hand off between them is a file on disk rather than a function call.

C.1 Implied volatility pipeline

`scripts/impl_vol/`

<code>data.py</code>	option chains from Yahoo Finance, cached by date
<code>rates.py</code>	US Treasury par curve, bootstrapped to zero rates
<code>implied_vol.py</code>	Black-Scholes prices, vega, implied vol inversion
<code>main.py</code>	the implied volatility surface and smile slices
<code>diag_naive_q.py</code>	the three stage dividend yield calibration
<code>export_smiles.py</code>	exports the per expiration call price smiles

The dependency flow is linear. `data.py` and `rates.py` fetch and cache the raw inputs, `implied_vol.py` provides the pricing and inversion used by everything else, and `main.py`,

diag_naive_q.py and export_smiles.py are the three entry points. The methodology behind them is Appendix A.

C.2 Local volatility calibration

```
scripts/local_vol_egger/  
  pde/  
    forward.py      forward Dupire solve, implicit Euler, P1 elements  
    linearized.py   linearized (sensitivity) solve, one direction h  
    adjoint.py      backward adjoint solve, accumulates the gradient  
  optimization/  
    vol.py          the coefficient  $a(y, \tau)$  as a stack of spatial slices  
    tikhonov.py     functional, gradient, and the L-BFGS-B driver  
    surface.py      the time dependent extension, one slice per maturity  
  utils.py          Black-Scholes helpers, meshes, smile handling  
  derivative_test.py gradient checks (duality and Taylor)  
  main.py           entry point, selects and runs one example  
  examples/  
    example1.py     Example 1, cases A to D  
    example2.py     Example 2, oscillatory coefficient  
    example3_market.py single maturity fit on market data  
    example4_surface.py the calibrated market surface  
    example5_skew.py local vs implied skew, factor of two  
    example6_holdout.py the out of sample pricing test  
    example7_arbitrage.py density and calendar arbitrage check  
    example8_discounting.py the two normalizations of  $u$ 
```

The three modules in pde/ are the three parabolic problems of Section 7.5. optimization/tikhonov.py assembles the functional (7.6) and calls pde/forward.py and pde/adjoint.py once per iteration to obtain the gradient (7.9). pde/linearized.py is not used in the calibration itself, it exists so that derivative_test.py can check the adjoint against an independent computation of the same directional derivative. optimization/surface.py wraps the same machinery for the stack of slices (7.16) and minimizes (7.17). Each file in examples/ is self contained and writes its own figures.

C.3 Where each result comes from

Result	Script
Implied volatility surface and smiles, Chapter 4	<code>impl_vol/main.py</code>
Dividend yield calibration, Appendix A	<code>impl_vol/diag_naive_q.py</code>
Call price smiles used as data	<code>impl_vol/export_smiles.py</code>
Gradient verification, Table 7.1	<code>local_vol_egger/derivative_test.py</code>
Example 1, Figures 7.1 to 7.3	<code>examples/example1.py</code>
Example 2, Figure 7.4	<code>examples/example2.py</code>
Market surface, Figures 7.6 to 7.8	<code>examples/example4_surface.py</code>
No-arbitrage check, Figure 7.5	<code>examples/example7_arbitrage.py</code>
Factor of two, Figure 7.9 and Table 7.3	<code>examples/example5_skew.py</code>
Out of sample test, Figure 7.10	<code>examples/example6_holdout.py</code>
Discounted normalization, Appendix B.12	<code>examples/example8_discounting.py</code>

Table C.1: Each reported result and the script that produces it.

BIBLIOGRAPHY

- [1] I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime, and G. N. Wells.
Dolfinx: The next generation fenics problem solving environment.
2023.
doi: 10.5281/zenodo.10447666.
- [2] H. Berestycki, J. Busca, and I. Florent.
Asymptotics and calibration of local volatility models.
Quantitative Finance, 2(1):61–69, 2002.
- [3] L. Bergomi.
Stochastic Volatility Modeling.
Chapman and Hall/CRC Financial Mathematics Series. CRC Press, 2015.
ISBN 9781482244076.
URL <https://books.google.es/books?id=dzs0CwAAQBAJ>.
- [4] T. Björk.
Arbitrage Theory in Continuous Time.
Oxford Finance. Oxford University Press, 2009.
ISBN 9780199574742.
URL <https://books.google.es/books?id=qEGDj5-BJ80C>.
- [5] Fischer Black and Myron S. Scholes.
The pricing of options and corporate liabilities.
Journal of Political Economy, 1973.
- [6] D.T. Breeden and R.H. Litzenberger.
Prices of state-contingent claims implicit in option prices.
Journal of Business, 51(4):621–651, 1978.
- [7] H. Brezis.
Functional Analysis, Sobolev Spaces and Partial Differential Equations.
Universitext. Springer New York, 2011.
ISBN 9780387709130.

- [8] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu.
A limited memory algorithm for bound constrained optimization.
SIAM Journal on Scientific Computing, 16(5):1190–1208, 1995.
- [9] E. Derman, M.B. Miller, and D. Park.
The Volatility Smile.
Wiley Finance. Wiley, 2016.
ISBN 9781118959169.
URL <https://books.google.es/books?id=WKfBDAAAQBAJ>.
- [10] B. Dupire.
Pricing with a smile.
Risk, 7(1):18–20, 1994.
- [11] Herbert Egger and Heinz W. Engl.
Tikhonov regularization applied to the inverse problem of option pricing: convergence analysis and rates.
Inverse Problems, 21(3):1027–1045, 2005.
doi: 10.1088/0266-5611/21/3/014.
- [12] Heinz W. Engl, Martin Hanke, and Andreas Neubauer.
Regularization of Inverse Problems, volume 375 of *Mathematics and Its Applications*.
Kluwer Academic Publishers, Dordrecht, 1996.
- [13] F. G. Friedlander and M. Joshi.
Introduction to the Theory of Distributions.
Cambridge University Press, 2 edition, 1998.
ISBN 9780521649711.
- [14] Michael B. Giles and Rebecca Carter.
Convergence analysis of crank-nicolson and rannacher time-marching.
Journal of Computational Finance, 9(4):89–112, 2006.
- [15] P. Glasserman.
Monte Carlo Methods in Financial Engineering.
Stochastic Modelling and Applied Probability. Springer New York, 2003.
ISBN 9780387004518.
- [16] O. Kallenberg.
Foundations of Modern Probability.
Probability and Its Applications. Springer, 2021.
URL <https://books.google.es/books?id=2ZY0swEACAAJ>.

- [17] A. Klenke.
Probability Theory: A Comprehensive Course.
Universitext. Springer, 2020.
URL <https://books.google.es/books?id=tcm3y5UJxDsC>.
- [18] Robert C. Merton.
Theory of rational option pricing.
Bell Journal of Economics and Management Science, 1973.
- [19] P. Mörters and Y. Peres.
Brownian Motion.
Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press,
2010.
ISBN 9781139486576.
URL <https://books.google.es/books?id=e-TbA-dSrZYC>.
- [20] B. Øksendal.
Stochastic Differential Equations: An Introduction with Applications.
Universitext. Springer Berlin Heidelberg, 2010.
ISBN 9783642143946.
URL <https://books.google.es/books?id=EQZEAAAQBAJ>.
- [21] J.M. Steele.
Stochastic Calculus and Financial Applications.
Stochastic Modelling and Applied Probability. Springer New York, 2010.
ISBN 9781441928627.
URL <https://books.google.es/books?id=bDmqcQAACAAJ>.
- [22] Vidar Thomée.
Galerkin Finite Element Methods for Parabolic Problems, volume 25 of *Springer Series in Computational Mathematics*.
Springer, Berlin, 2 edition, 2006.